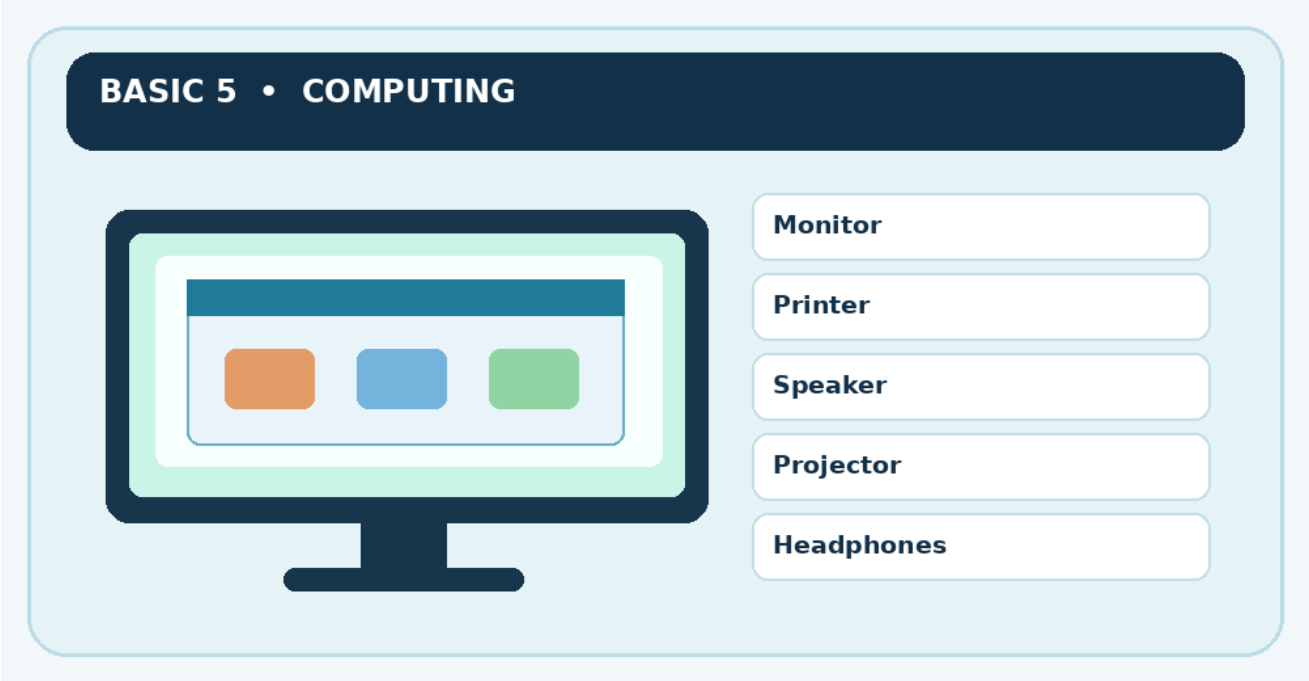
COMPUTING

Basic 5 — Illustrated Learning Workbook

Learn • Explore • Practise • Play • Challenge



34 lessons | 170 Play questions | 340 Challenge questions

Basic 4–6: independent adaptation of selected indicators in the 2019 NaCCA Computing curriculum; not an exhaustive mapping.

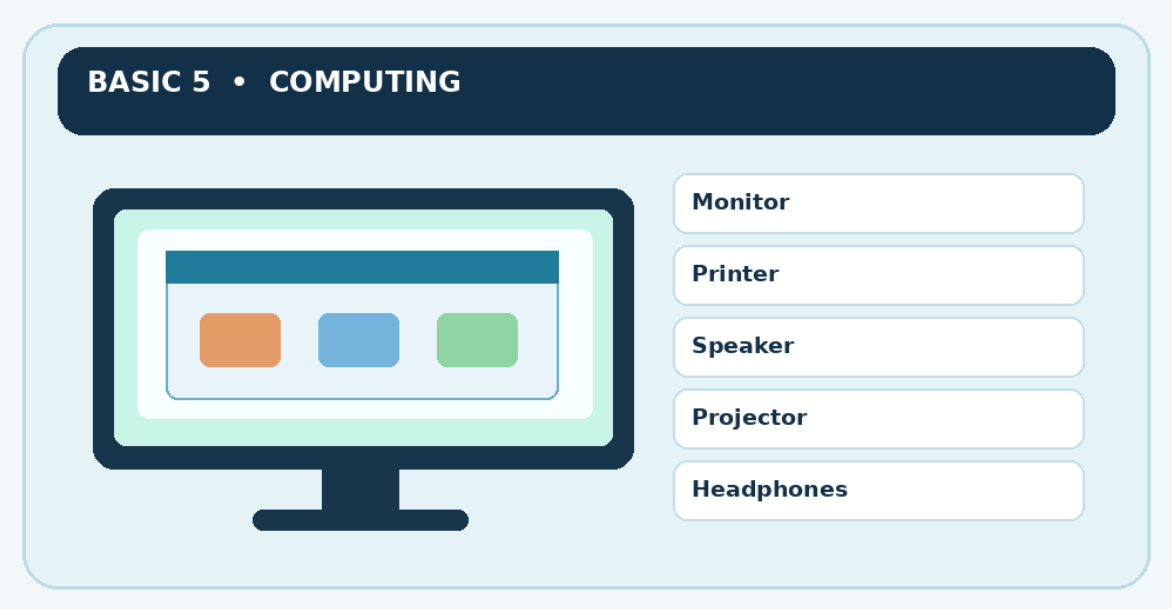
Course guide and lesson list

Start with the picture and five important ideas. Read the Ghanaian example, complete the practice activity and answer the quizzes. An adult should supervise device use and online activities.

01. Recognising Output Devices — Computing Systems
02. Uses of Output Devices — Computing Systems
03. Analogue and Digital Devices — Computing Systems
04. Advanced Mouse Practice — Computing Systems
05. Keyboard Rows and Short Text — Computing Systems
06. Windows Desktop Interface — Computer Applications
07. How Computers Have Changed — Computing Systems
08. Create a Desktop Background — Digital Creation
09. Start Menu and Pinned Apps — Computer Applications
10. Desktop Shortcuts — Computer Applications
11. Move and Copy Files — Computer Applications
12. Use File Explorer — Computer Applications
13. Data From Experiments — Data and Information
14. Primary Sources of Data — Data and Information
15. Collect a Data Sample — Data and Information
16. Analyse Charts and Tables — Data and Information
17. How Technology Affects Communities — Technology in Society
18. Hardware, Software and Networks — Computing Systems
19. PowerPoint Design and Animation — Presentation
20. Present Five Slides — Presentation
21. Word Processing Ribbons — Word Processing
22. Design a Well-Formatted Document — Digital Creation
23. Introducing Databases — Programming and Databases
24. Why We Use Databases — Programming and Databases
25. Database Tables and Records — Programming and Databases
26. Programming Languages — Programming and Databases
27. Understanding Web Browsers — Internet and Communication
28. What Is a Web Page? — Internet and Communication
29. Following Hyperlinks — Internet and Communication
30. Email Services — Internet and Communication
31. Internet of Things — Technology in Society
32. Your Digital Footprint — Digital Citizenship
33. Computer Health Hazards — Health and Safety
34. Healthy Computer Posture — Health and Safety

Lesson 1: Recognising Output Devices

We will learn to: Identify, explain and apply the important ideas behind recognising output devices.



Five important ideas

Monitor: displays visual output.

Printer: makes output on paper.

Speaker: produces audio output.

Projector: shows images to a group.

Headphones: provide sound for one listener.

Our Ghanaian story

During a class presentation, Kojo had a question about recognising output devices. The group discovered that monitor displays visual output, while printer makes output on paper. They drew an example and explained it to their classmates.

Try it!

In a small group, draw or safely examine pictures of monitor, printer and speaker. Explain one real-life use for each in Ghana.

Create and share

Make a colourful, labelled poster that teaches recognising output devices. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.1.1.1

Lesson 1: Play & Challenge

PLAY — Choose one best answer for each question.

1. Recognising Output Devices — Choose the correct name for something that displays visual output.

A. Speaker B. Headphones C. Projector D. Monitor

2. Recognising Output Devices — Your friend says, "This makes output on paper." What is the friend describing?

A. Monitor B. Projector C. Printer D. Headphones

3. Recognising Output Devices — Which idea from recognising output devices matches the clue "produces audio output"?

A. Printer B. Headphones C. Monitor D. Speaker

4. Recognising Output Devices — The teacher asks for an example that shows images to a group. Which answer fits?

A. Monitor B. Headphones C. Printer D. Projector

5. Recognising Output Devices — Can you identify the term described here: provide sound for one listener?

A. Speaker B. Monitor C. Projector D. Headphones

CHALLENGE — Think and apply.

1. Recognising Output Devices — A classmate confuses several ideas about recognising output devices. Which one actually makes output on paper?

A. Projector B. Printer C. Headphones D. Monitor

2. Recognising Output Devices — At home, Yaw remembers a class rule or concept that shows images to a group. What was it called?

A. Monitor B. Printer C. Headphones D. Projector

3. Recognising Output Devices — Imagine helping a younger pupil understand something that produces audio output. Which term would you teach?

A. Monitor B. Headphones C. Printer D. Speaker

4. Recognising Output Devices — For a short demonstration, the teacher needs an example that produces audio output. Which choice would you select?

A. Speaker B. Monitor C. Headphones D. Printer

5. Recognising Output Devices — A group makes a display on recognising output devices. Their first clue is "makes output on paper". Which label is correct?

A. Printer B. Monitor C. Headphones D. Projector

6. Recognising Output Devices — During a school ICT lesson, Yaw wants to show something that displays visual output. Which idea should be labelled?

A. Headphones B. Monitor C. Projector D. Speaker

7. Recognising Output Devices — Which answer would fit on a poster beside the words "displays visual output"?

A. Monitor B. Projector C. Speaker D. Headphones

8. Recognising Output Devices — Your class is solving a practical problem. One useful thing shows images to a group. What is its name?

A. Projector B. Monitor C. Printer D. Headphones

9. Recognising Output Devices — At the community learning centre, someone describes a tool or idea that provide sound for one listener. Which one is it?

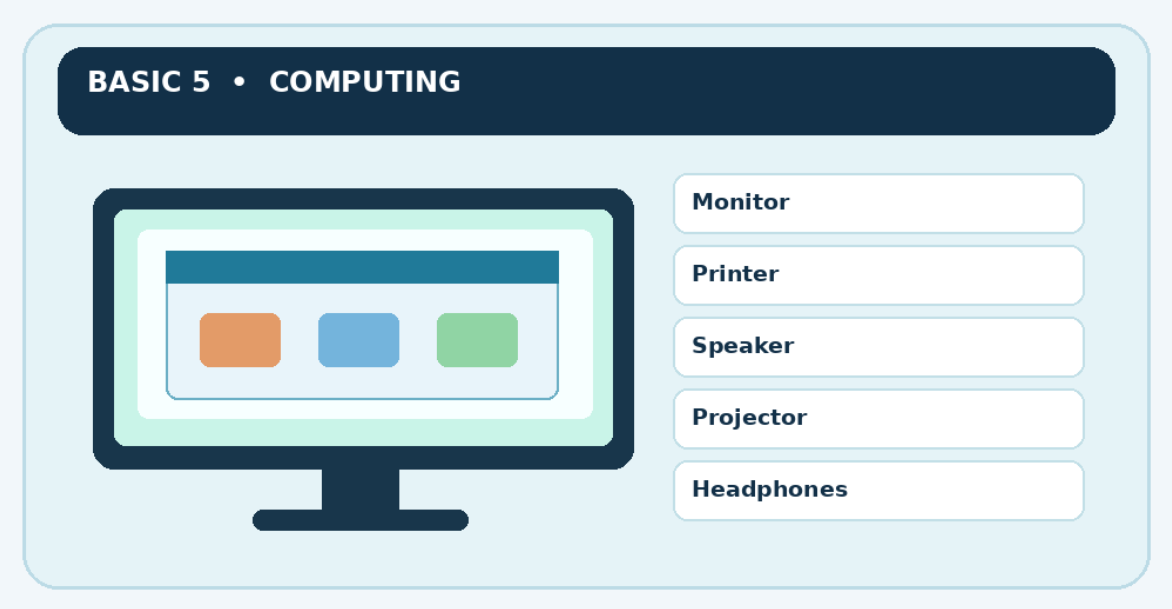
A. Monitor B. Projector C. Speaker D. Headphones

10. Recognising Output Devices — Which concept matches the explanation "provide sound for one listener" when you review recognising output devices?

A. Headphones B. Monitor C. Speaker D. Projector

Lesson 2: Uses of Output Devices

We will learn to: Identify, explain and apply the important ideas behind uses of output devices.



Five important ideas

Monitor: displays visual output.

Printer: makes output on paper.

Speaker: produces audio output.

Projector: shows images to a group.

Headphones: provide sound for one listener.

Our Ghanaian story

During a class presentation, Esi had a question about uses of output devices. The group discovered that monitor displays visual output, while printer makes output on paper. They drew an example and explained it to their classmates.

Try it!

In a small group, draw or safely examine pictures of monitor, printer and speaker. Explain one real-life use for each in Ghana.

Create and share

Make a colourful, labelled poster that teaches uses of output devices. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.1.1.2

Lesson 2: Play & Challenge

PLAY — Choose one best answer for each question.

1. Uses of Output Devices — Choose the correct name for something that displays visual output.

A. Printer B. Projector C. Headphones D. Monitor

2. Uses of Output Devices — Your friend says, "This makes output on paper." What is the friend describing?

A. Printer B. Monitor C. Projector D. Speaker

3. Uses of Output Devices — Which idea from uses of output devices matches the clue "produces audio output"?

A. Printer B. Projector C. Monitor D. Speaker

4. Uses of Output Devices — The teacher asks for an example that shows images to a group. Which answer fits?

A. Projector B. Monitor C. Headphones D. Speaker

5. Uses of Output Devices — Can you identify the term described here: provide sound for one listener?

A. Headphones B. Printer C. Projector D. Speaker

CHALLENGE — Think and apply.

1. Uses of Output Devices — During a school ICT lesson, Kofi wants to show something that displays visual output. Which idea should be labelled?

A. Projector B. Printer C. Headphones D. Monitor

2. Uses of Output Devices — A group makes a display on uses of output devices. Their first clue is "makes output on paper". Which label is correct?

A. Printer B. Speaker C. Monitor D. Projector

3. Uses of Output Devices — Your class is solving a practical problem. One useful thing shows images to a group. What is its name?

A. Projector B. Monitor C. Headphones D. Speaker

4. Uses of Output Devices — For a short demonstration, the teacher needs an example that produces audio output. Which choice would you select?

A. Printer B. Speaker C. Projector D. Monitor

5. Uses of Output Devices — Which concept matches the explanation "provide sound for one listener" when you review uses of output devices?

A. Projector B. Headphones C. Printer D. Speaker

6. Uses of Output Devices — A classmate confuses several ideas about uses of output devices. Which one actually makes output on paper?

A. Projector B. Printer C. Speaker D. Monitor

7. Uses of Output Devices — At home, Kofi remembers a class rule or concept that shows images to a group. What was it called?

A. Headphones B. Speaker C. Projector D. Monitor

8. Uses of Output Devices — At the community learning centre, someone describes a tool or idea that provide sound for one listener. Which one is it?

A. Projector B. Printer C. Headphones D. Speaker

9. Uses of Output Devices — Imagine helping a younger pupil understand something that produces audio output. Which term would you teach?

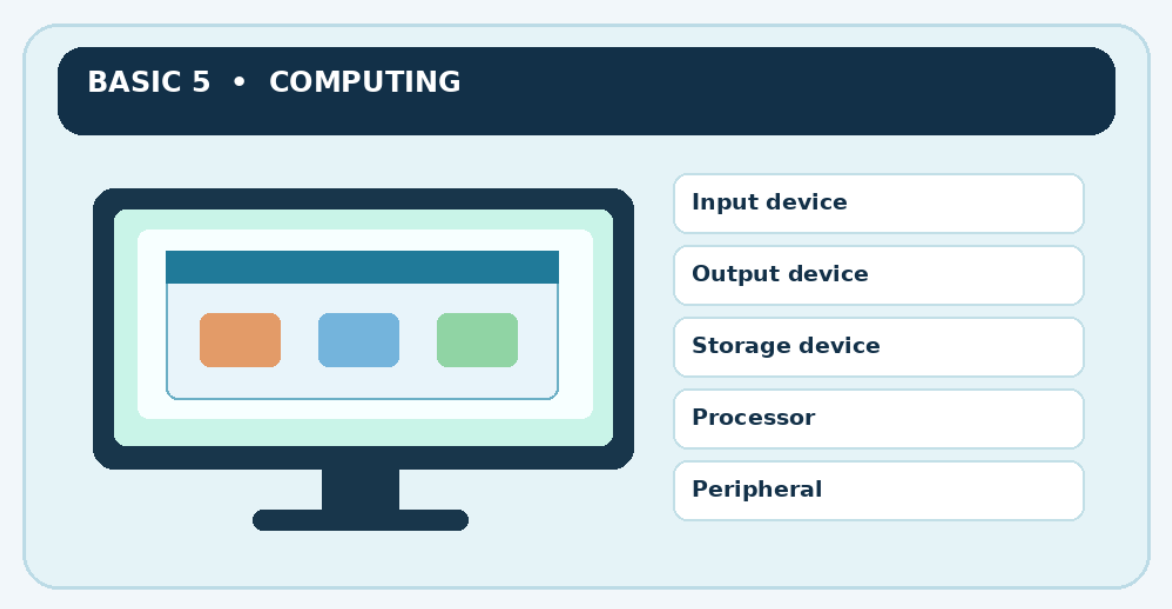
A. Monitor B. Printer C. Speaker D. Projector

10. Uses of Output Devices — Which answer would fit on a poster beside the words "displays visual output"?

A. Printer B. Monitor C. Projector D. Headphones

Lesson 3: Analogue and Digital Devices

We will learn to: Identify, explain and apply the important ideas behind analogue and digital devices.



Five important ideas

Input device: sends information into a computer.

Output device: shows or plays results.

Storage device: keeps digital files.

Processor: helps follow computing instructions.

Peripheral: is an extra device connected to a computer.

Our Ghanaian story

During a tour of the computer room, Yaw had a question about analogue and digital devices. The group discovered that input device sends information into a computer, while output device shows or plays results. They drew an example and explained it to their classmates.

Try it!

In a small group, draw or safely examine pictures of input device, output device and storage device. Explain one real-life use for each in Ghana.

Create and share

Make a colourful, labelled poster that teaches analogue and digital devices. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.1.1.3

Lesson 3: Play & Challenge

PLAY — Choose one best answer for each question.

1. Analogue and Digital Devices — Choose the correct name for something that sends information into a computer.

A. Output device B. Peripheral C. Input device D. Storage device

2. Analogue and Digital Devices — Your friend says, "This shows or plays results." What is the friend describing?

A. Output device B. Peripheral C. Storage device D. Processor

3. Analogue and Digital Devices — Which idea from analogue and digital devices matches the clue "keeps digital files"?

A. Storage device B. Input device C. Output device D. Processor

4. Analogue and Digital Devices — The teacher asks for an example that helps follow computing instructions. Which answer fits?

A. Input device B. Storage device C. Processor D. Peripheral

5. Analogue and Digital Devices — Can you identify the term described here: is an extra device connected to a computer?

A. Peripheral B. Processor C. Storage device D. Input device

CHALLENGE — Think and apply.

1. Analogue and Digital Devices — Imagine helping a younger pupil understand something that keeps digital files. Which term would you teach?

A. Processor B. Storage device C. Output device D. Input device

2. Analogue and Digital Devices — Which answer would fit on a poster beside the words "sends information into a computer"?

A. Input device B. Peripheral C. Storage device D. Output device

3. Analogue and Digital Devices — Your class is solving a practical problem. One useful thing helps follow computing instructions. What is its name?

A. Storage device B. Peripheral C. Input device D. Processor

4. Analogue and Digital Devices — During a school ICT lesson, Akosua wants to show something that sends information into a computer. Which idea should be labelled?

A. Peripheral B. Input device C. Storage device D. Output device

5. Analogue and Digital Devices — A group makes a display on analogue and digital devices. Their first clue is "shows or plays results". Which label is correct?

A. Peripheral B. Output device C. Storage device D. Processor

6. Analogue and Digital Devices — At home, Akosua remembers a class rule or concept that helps follow computing instructions. What was it called?

A. Storage device B. Processor C. Input device D. Peripheral

7. Analogue and Digital Devices — For a short demonstration, the teacher needs an example that keeps digital files. Which choice would you select?

A. Input device B. Storage device C. Processor D. Output device

8. Analogue and Digital Devices — A classmate confuses several ideas about analogue and digital devices. Which one actually shows or plays results?

A. Storage device B. Output device C. Peripheral D. Processor

9. Analogue and Digital Devices — Which concept matches the explanation "is an extra device connected to a computer" when you review analogue and digital devices?

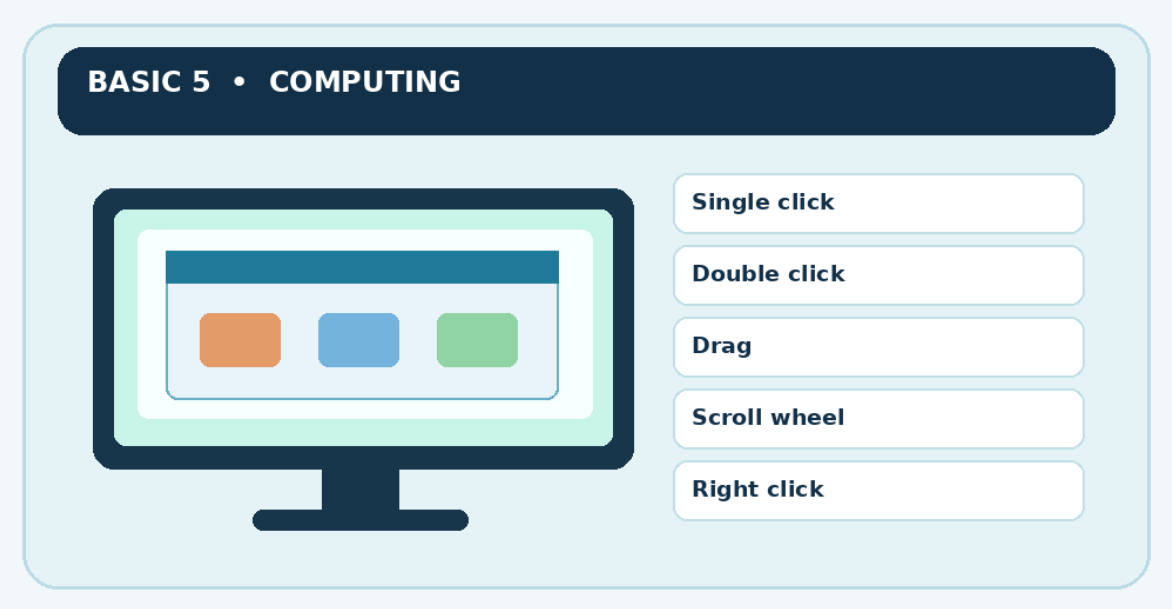
A. Input device B. Storage device C. Peripheral D. Processor

10. Analogue and Digital Devices — At the community learning centre, someone describes a tool or idea that is an extra device connected to a computer. Which one is it?

A. Processor B. Peripheral C. Storage device D. Input device

Lesson 4: Advanced Mouse Practice

We will learn to: Identify, explain and apply the important ideas behind advanced mouse practice.



Five important ideas

Single click: selects an item.

Double click: opens many files or apps.

Drag: holds the button while moving an item.

Scroll wheel: moves through a page.

Right click: opens a context menu in many systems.

Our Ghanaian story

At a shared classroom computer, Abena had a question about advanced mouse practice. The group discovered that single click selects an item, while double click opens many files or apps. They drew an example and explained it to their classmates.

Try it!

With an adult supervising, practise single click and double click using a computer if one is available. Without a computer, draw the screen and explain each step to a partner.

Create and share

Make a colourful, labelled poster that teaches advanced mouse practice. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.1.1.4

Lesson 4: Play & Challenge

PLAY — Choose one best answer for each question.

1. Advanced Mouse Practice — Choose the correct name for something that selects an item.

A. Scroll wheel B. Right click C. Single click D. Drag

2. Advanced Mouse Practice — Your friend says, "This opens many files or apps." What is the friend describing?

A. Double click B. Right click C. Single click D. Scroll wheel

3. Advanced Mouse Practice — Which idea from advanced mouse practice matches the clue "holds the button while moving an item"?

A. Double click B. Scroll wheel C. Right click D. Drag

4. Advanced Mouse Practice — The teacher asks for an example that moves through a page. Which answer fits?

A. Double click B. Single click C. Right click D. Scroll wheel

5. Advanced Mouse Practice — Can you identify the term described here: opens a context menu in many systems?

A. Double click B. Scroll wheel C. Right click D. Drag

CHALLENGE — Think and apply.

1. Advanced Mouse Practice — Imagine helping a younger pupil understand something that holds the button while moving an item. Which term would you teach?

A. Right click B. Scroll wheel C. Double click D. Drag

2. Advanced Mouse Practice — A classmate confuses several ideas about advanced mouse practice. Which one actually opens many files or apps?

A. Single click B. Right click C. Double click D. Scroll wheel

3. Advanced Mouse Practice — At the community learning centre, someone describes a tool or idea that opens a context menu in many systems. Which one is it?

A. Drag B. Double click C. Scroll wheel D. Right click

4. Advanced Mouse Practice — For a short demonstration, the teacher needs an example that holds the button while moving an item. Which choice would you select?

A. Scroll wheel B. Double click C. Right click D. Drag

5. Advanced Mouse Practice — During a school ICT lesson, Esi wants to show something that selects an item. Which idea should be labelled?

A. Scroll wheel B. Drag C. Single click D. Right click

6. Advanced Mouse Practice — Your class is solving a practical problem. One useful thing moves through a page. What is its name?

A. Double click B. Scroll wheel C. Single click D. Right click

7. Advanced Mouse Practice — At home, Esi remembers a class rule or concept that moves through a page. What was it called?

A. Single click B. Right click C. Scroll wheel D. Double click

8. Advanced Mouse Practice — Which concept matches the explanation "opens a context menu in many systems" when you review advanced mouse practice?

A. Right click B. Double click C. Drag D. Scroll wheel

9. Advanced Mouse Practice — Which answer would fit on a poster beside the words "selects an item"?

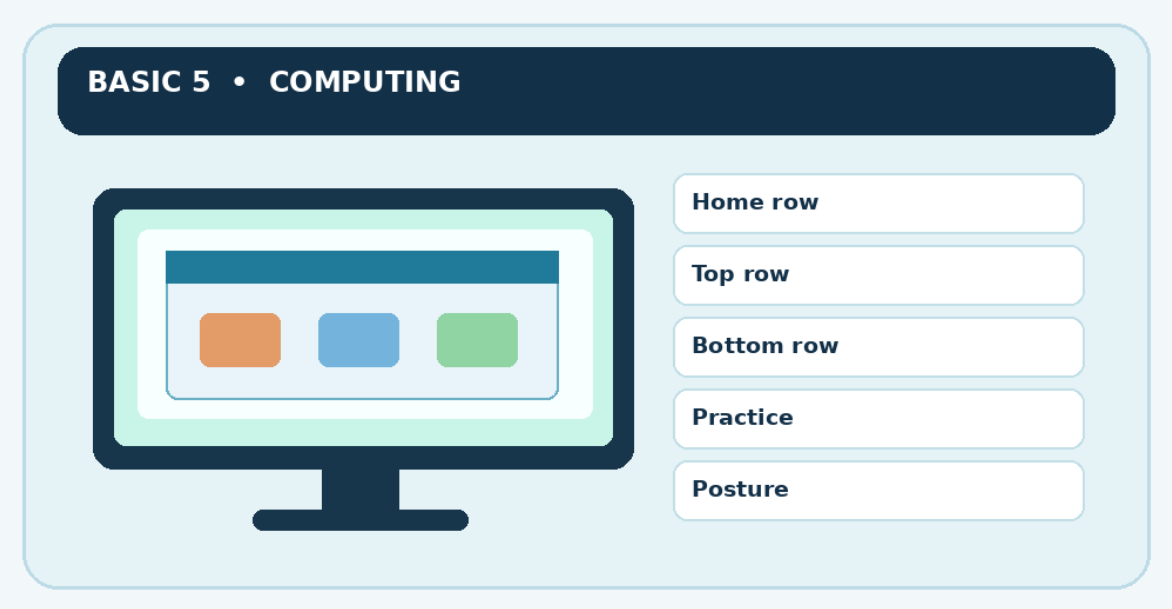
A. Scroll wheel B. Right click C. Single click D. Drag

10. Advanced Mouse Practice — A group makes a display on advanced mouse practice. Their first clue is "opens many files or apps". Which label is correct?

A. Double click B. Right click C. Scroll wheel D. Single click

Lesson 5: Keyboard Rows and Short Text

We will learn to: Identify, explain and apply the important ideas behind keyboard rows and short text.



Five important ideas

Home row: is the resting row for touch typing.

Top row: contains letters Q through P.

Bottom row: contains letters Z through M.

Practice: helps develop accuracy before speed.

Posture: keeps shoulders relaxed while typing.

Our Ghanaian story

At an ICT typing competition, Kofi had a question about keyboard rows and short text. The group discovered that home row is the resting row for touch typing, while top row contains letters Q through P. They drew an example and explained it to their classmates.

Try it!

With an adult supervising, practise home row and top row using a computer if one is available. Without a computer, draw the screen and explain each step to a partner.

Create and share

Make a colourful, labelled poster that teaches keyboard rows and short text. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.1.1.5

Lesson 5: Play & Challenge

PLAY — Choose one best answer for each question.

1. Keyboard Rows and Short Text — Choose the correct name for something that is the resting row for touch typing.

A. Practice B. Top row C. Posture D. Home row

2. Keyboard Rows and Short Text — Your friend says, "This contains letters Q through P." What is the friend describing?

A. Top row B. Home row C. Practice D. Bottom row

3. Keyboard Rows and Short Text — Which idea from keyboard rows and short text matches the clue "contains letters Z through M"?

A. Posture B. Home row C. Top row D. Bottom row

4. Keyboard Rows and Short Text — The teacher asks for an example that helps develop accuracy before speed. Which answer fits?

A. Top row B. Home row C. Posture D. Practice

5. Keyboard Rows and Short Text — Can you identify the term described here: keeps shoulders relaxed while typing?

A. Practice B. Posture C. Top row D. Bottom row

CHALLENGE — Think and apply.

1. Keyboard Rows and Short Text — A group makes a display on keyboard rows and short text. Their first clue is "contains letters Q through P". Which label is correct?

A. Practice B. Home row C. Bottom row D. Top row

2. Keyboard Rows and Short Text — For a short demonstration, the teacher needs an example that contains letters Z through M. Which choice would you select?

A. Home row B. Posture C. Top row D. Bottom row

3. Keyboard Rows and Short Text — Which concept matches the explanation "keeps shoulders relaxed while typing" when you review keyboard rows and short text?

A. Posture B. Top row C. Bottom row D. Practice

4. Keyboard Rows and Short Text — At home, Ama remembers a class rule or concept that helps develop accuracy before speed. What was it called?

A. Posture B. Home row C. Top row D. Practice

5. Keyboard Rows and Short Text — A classmate confuses several ideas about keyboard rows and short text. Which one actually contains letters Q through P?

A. Practice B. Top row C. Home row D. Bottom row

6. Keyboard Rows and Short Text — At the community learning centre, someone describes a tool or idea that keeps shoulders relaxed while typing. Which one is it?

A. Top row B. Posture C. Bottom row D. Practice

7. Keyboard Rows and Short Text — Your class is solving a practical problem. One useful thing helps develop accuracy before speed. What is its name?

A. Top row B. Posture C. Home row D. Practice

8. Keyboard Rows and Short Text — Which answer would fit on a poster beside the words "is the resting row for touch typing"?

A. Top row B. Posture C. Practice D. Home row

9. Keyboard Rows and Short Text — During a school ICT lesson, Ama wants to show something that is the resting row for touch typing. Which idea should be labelled?

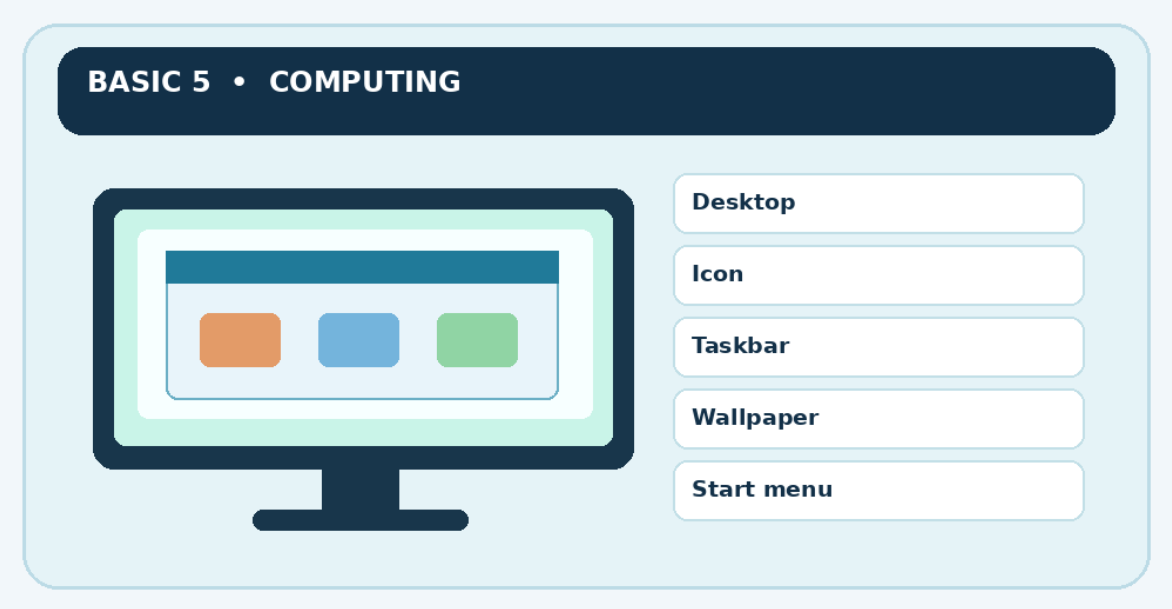
A. Practice B. Posture C. Home row D. Top row

10. Keyboard Rows and Short Text — Imagine helping a younger pupil understand something that contains letters Z through M. Which term would you teach?

A. Posture B. Home row C. Bottom row D. Top row

Lesson 6: Windows Desktop Interface

We will learn to: Identify, explain and apply the important ideas behind windows desktop interface.



Five important ideas

Desktop: is a main computer work area.

Icon: is a small picture representing an item.

Taskbar: helps find open apps and tools.

Wallpaper: is the desktop background picture.

Start menu: offers a way to open programs.

Our Ghanaian story

When the computer starts, Ama had a question about windows desktop interface. The group discovered that desktop is a main computer work area, while icon is a small picture representing an item. They drew an example and explained it to their classmates.

Try it!

With an adult supervising, practise desktop and icon using a computer if one is available. Without a computer, draw the screen and explain each step to a partner.

Create and share

Make a colourful, labelled poster that teaches windows desktop interface. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.1.1.6

Lesson 6: Play & Challenge

PLAY — Choose one best answer for each question.

1. Windows Desktop Interface — Choose the correct name for something that is a main computer work area.

A. Wallpaper B. Icon C. Desktop D. Taskbar

2. Windows Desktop Interface — Your friend says, "This is a small picture representing an item." What is the friend describing?

A. Wallpaper B. Taskbar C. Icon D. Desktop

3. Windows Desktop Interface — Which idea from windows desktop interface matches the clue "helps find open apps and tools"?

A. Wallpaper B. Icon C. Start menu D. Taskbar

4. Windows Desktop Interface — The teacher asks for an example that is the desktop background picture. Which answer fits?

A. Start menu B. Wallpaper C. Taskbar D. Desktop

5. Windows Desktop Interface — Can you identify the term described here: offers a way to open programs?

A. Start menu B. Desktop C. Taskbar D. Icon

CHALLENGE — Think and apply.

1. Windows Desktop Interface — For a short demonstration, the teacher needs an example that helps find open apps and tools. Which choice would you select?

A. Icon B. Start menu C. Wallpaper D. Taskbar

2. Windows Desktop Interface — At the community learning centre, someone describes a tool or idea that offers a way to open programs. Which one is it?

A. Icon B. Desktop C. Taskbar D. Start menu

3. Windows Desktop Interface — Your class is solving a practical problem. One useful thing is the desktop background picture. What is its name?

A. Taskbar B. Desktop C. Start menu D. Wallpaper

4. Windows Desktop Interface — Imagine helping a younger pupil understand something that helps find open apps and tools. Which term would you teach?

A. Taskbar B. Start menu C. Wallpaper D. Icon

5. Windows Desktop Interface — Which answer would fit on a poster beside the words "is a main computer work area"?

A. Desktop B. Taskbar C. Wallpaper D. Icon

6. Windows Desktop Interface — A group makes a display on windows desktop interface. Their first clue is "is a small picture representing an item". Which label is correct?

A. Taskbar B. Wallpaper C. Icon D. Desktop

7. Windows Desktop Interface — Which concept matches the explanation "offers a way to open programs" when you review windows desktop interface?

A. Start menu B. Icon C. Desktop D. Taskbar

8. Windows Desktop Interface — A classmate confuses several ideas about windows desktop interface. Which one actually is a small picture representing an item?

A. Icon B. Wallpaper C. Desktop D. Taskbar

9. Windows Desktop Interface — During a school ICT lesson, Yaw wants to show something that is a main computer work area. Which idea should be labelled?

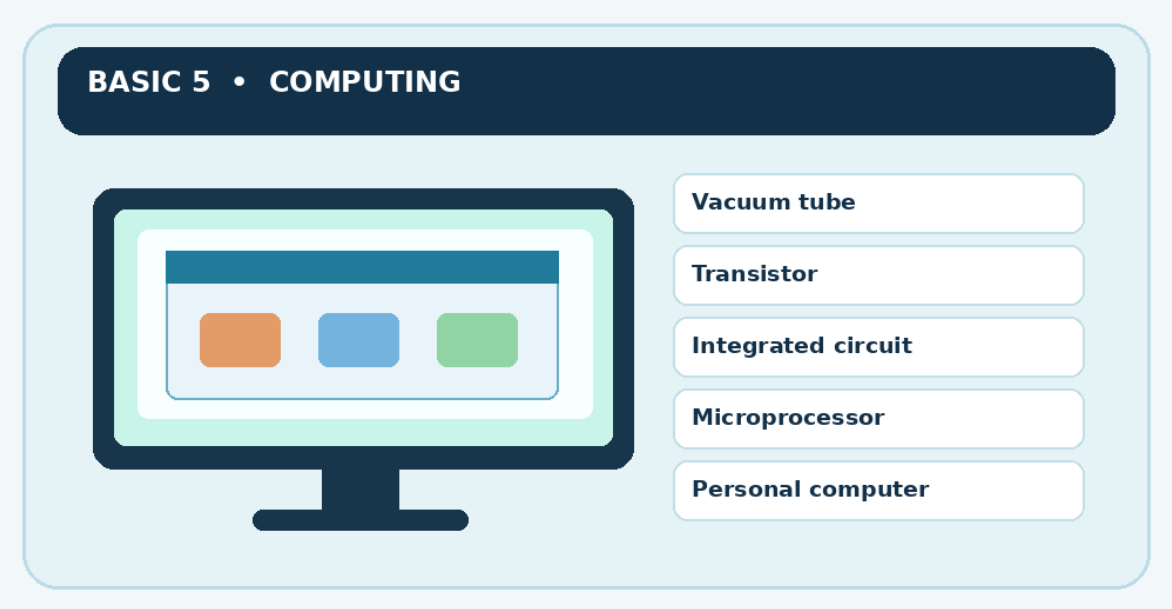
A. Taskbar B. Desktop C. Icon D. Wallpaper

10. Windows Desktop Interface — At home, Yaw remembers a class rule or concept that is the desktop background picture. What was it called?

A. Taskbar B. Desktop C. Wallpaper D. Start menu

Lesson 7: How Computers Have Changed

We will learn to: Identify, explain and apply the important ideas behind how computers have changed.



Five important ideas

Vacuum tube: was used in many first-generation computers.

Transistor: replaced vacuum tubes in many later machines.

Integrated circuit: combined electronic parts on a chip.

Microprocessor: puts processing circuitry onto a chip.

Personal computer: is designed for use by an individual.

Our Ghanaian story

While looking at old and new computer photographs, Kojo had a question about how computers have changed. The group discovered that vacuum tube was used in many first-generation computers, while transistor replaced vacuum tubes in many later machines. They drew an example and explained it to their classmates.

Try it!

In a small group, draw or safely examine pictures of vacuum tube, transistor and integrated circuit. Explain one real-life use for each in Ghana.

Create and share

Make a colourful, labelled poster that teaches how computers have changed. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.1.1.7

Lesson 7: Play & Challenge

PLAY — Choose one best answer for each question.

1. How Computers Have Changed — Choose the correct name for something that was used in many first-generation computers.

A. Transistor B. Microprocessor C. Personal computer D. Vacuum tube

2. How Computers Have Changed — Your friend says, "This replaced vacuum tubes in many later machines." What is the friend describing?

A. Vacuum tube B. Transistor C. Integrated circuit D. Microprocessor

3. How Computers Have Changed — Which idea from how computers have changed matches the clue "combined electronic parts on a chip"?

A. Microprocessor B. Personal computer C. Integrated circuit D. Transistor

4. How Computers Have Changed — The teacher asks for an example that puts processing circuitry onto a chip. Which answer fits?

A. Transistor B. Microprocessor C. Personal computer D. Vacuum tube

5. How Computers Have Changed — Can you identify the term described here: is designed for use by an individual?

A. Vacuum tube B. Personal computer C. Transistor D. Microprocessor

CHALLENGE — Think and apply.

1. How Computers Have Changed — Imagine helping a younger pupil understand something that combined electronic parts on a chip. Which term would you teach?

A. Personal computer B. Microprocessor C. Transistor D. Integrated circuit

2. How Computers Have Changed — Which concept matches the explanation "is designed for use by an individual" when you review how computers have changed?

A. Transistor B. Vacuum tube C. Microprocessor D. Personal computer

3. How Computers Have Changed — At the community learning centre, someone describes a tool or idea that is designed for use by an individual. Which one is it?

A. Microprocessor B. Vacuum tube C. Personal computer D. Transistor

4. How Computers Have Changed — At home, Kofi remembers a class rule or concept that puts processing circuitry onto a chip. What was it called?

A. Vacuum tube B. Transistor C. Personal computer D. Microprocessor

5. How Computers Have Changed — Which answer would fit on a poster beside the words "was used in many first-generation computers"?

A. Microprocessor B. Transistor C. Personal computer D. Vacuum tube

6. How Computers Have Changed — A classmate confuses several ideas about how computers have changed. Which one actually replaced vacuum tubes in many later machines?

A. Microprocessor B. Vacuum tube C. Integrated circuit D. Transistor

7. How Computers Have Changed — A group makes a display on how computers have changed. Their first clue is "replaced vacuum tubes in many later machines". Which label is correct?

A. Integrated circuit B. Transistor C. Microprocessor D. Vacuum tube

8. How Computers Have Changed — For a short demonstration, the teacher needs an example that combined electronic parts on a chip. Which choice would you select?

A. Personal computer B. Transistor C. Integrated circuit D. Microprocessor

9. How Computers Have Changed — Your class is solving a practical problem. One useful thing puts processing circuitry onto a chip. What is its name?

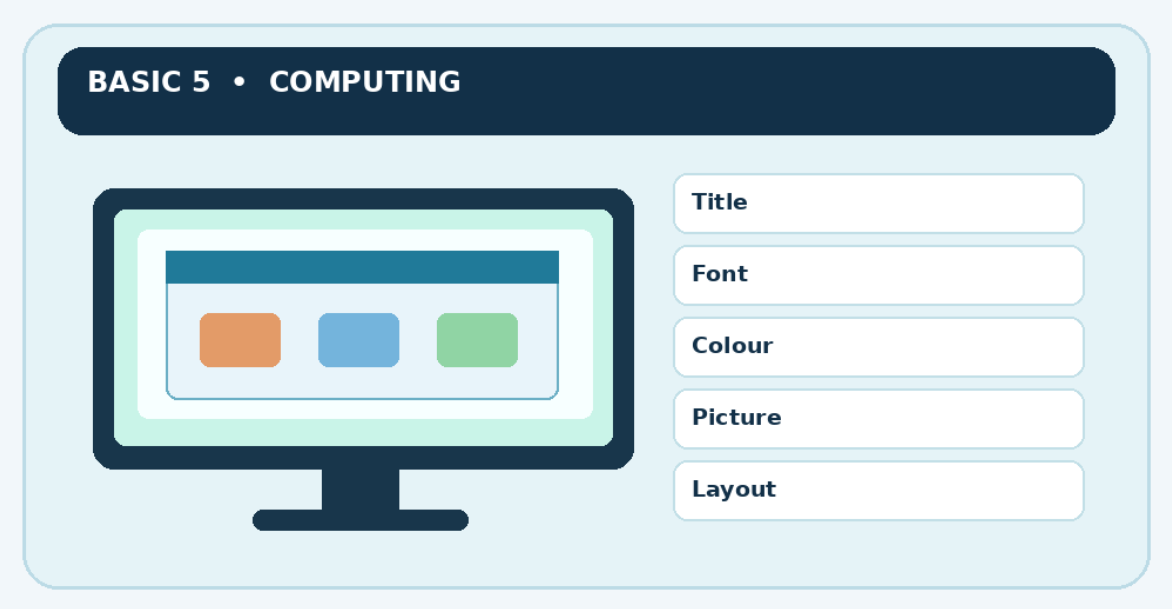
A. Personal computer B. Microprocessor C. Vacuum tube D. Transistor

10. How Computers Have Changed — During a school ICT lesson, Kofi wants to show something that was used in many first-generation computers. Which idea should be labelled?

A. Vacuum tube B. Transistor C. Personal computer D. Microprocessor

Lesson 8: Create a Desktop Background

We will learn to: Identify, explain and apply the important ideas behind create a desktop background.



Five important ideas

Title: states the topic of a page or slide.

Font: changes how writing appears.

Colour: helps emphasise important information.

Picture: can explain an idea visually.

Layout: arranges content clearly.

Our Ghanaian story

While making an Independence Day poster, Esi had a question about create a desktop background. The group discovered that title states the topic of a page or slide, while font changes how writing appears. They drew an example and explained it to their classmates.

Try it!

Make a one-page paper example about your school or a Ghanaian festival. Include title, font and colour; present it clearly to a partner.

Create and share

Make a colourful, labelled poster that teaches create a desktop background. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.2.1.1

Lesson 8: Play & Challenge

PLAY — Choose one best answer for each question.

1. Create a Desktop Background — Choose the correct name for something that states the topic of a page or slide.

A. Picture B. Title C. Layout D. Font

2. Create a Desktop Background — Your friend says, "This changes how writing appears." What is the friend describing?

A. Title B. Layout C. Font D. Picture

3. Create a Desktop Background — Which idea from create a desktop background matches the clue "helps emphasise important information"?

A. Layout B. Picture C. Colour D. Title

4. Create a Desktop Background — The teacher asks for an example that can explain an idea visually. Which answer fits?

A. Layout B. Picture C. Title D. Font

5. Create a Desktop Background — Can you identify the term described here: arranges content clearly?

A. Font B. Title C. Layout D. Colour

CHALLENGE — Think and apply.

1. Create a Desktop Background — For a short demonstration, the teacher needs an example that helps emphasise important information. Which choice would you select?

A. Colour B. Picture C. Title D. Layout

2. Create a Desktop Background — Which answer would fit on a poster beside the words "states the topic of a page or slide"?

A. Font B. Title C. Layout D. Picture

3. Create a Desktop Background — A classmate confuses several ideas about create a desktop background. Which one actually changes how writing appears?

A. Title B. Picture C. Font D. Layout

4. Create a Desktop Background — At home, Akosua remembers a class rule or concept that can explain an idea visually. What was it called?

A. Font B. Title C. Picture D. Layout

5. Create a Desktop Background — Your class is solving a practical problem. One useful thing can explain an idea visually. What is its name?

A. Picture B. Title C. Font D. Layout

6. Create a Desktop Background — A group makes a display on create a desktop background. Their first clue is "changes how writing appears". Which label is correct?

A. Layout B. Font C. Title D. Picture

7. Create a Desktop Background — At the community learning centre, someone describes a tool or idea that arranges content clearly. Which one is it?

A. Font B. Title C. Colour D. Layout

8. Create a Desktop Background — Which concept matches the explanation "arranges content clearly" when you review create a desktop background?

A. Colour B. Font C. Layout D. Title

9. Create a Desktop Background — Imagine helping a younger pupil understand something that helps emphasise important information. Which term would you teach?

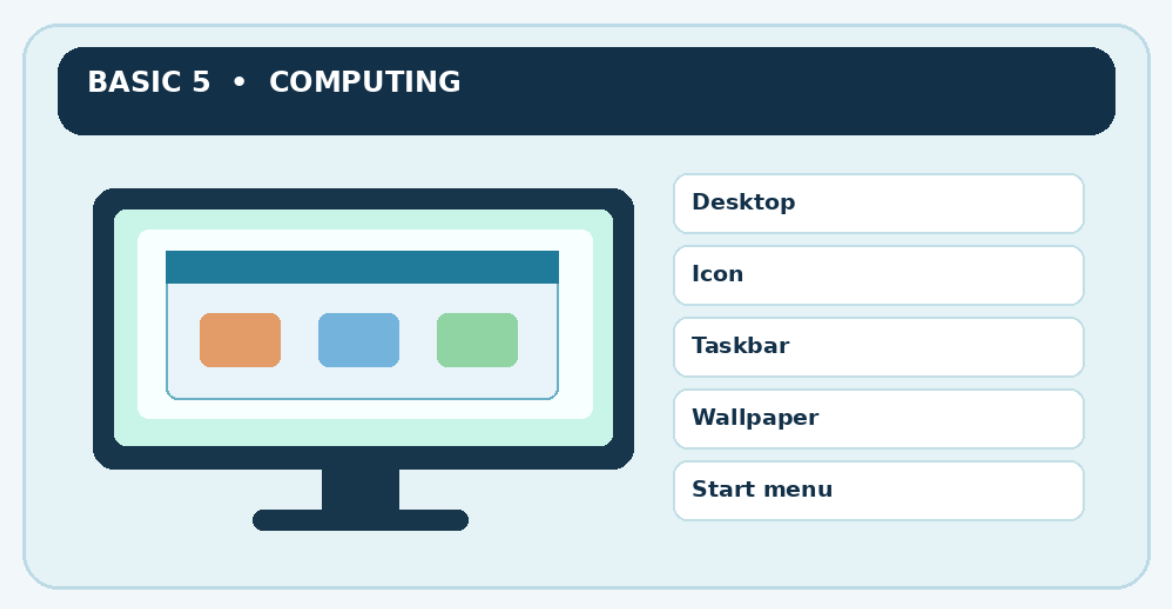
A. Title B. Picture C. Layout D. Colour

10. Create a Desktop Background — During a school ICT lesson, Akosua wants to show something that states the topic of a page or slide. Which idea should be labelled?

A. Title B. Layout C. Picture D. Font

Lesson 9: Start Menu and Pinned Apps

We will learn to: Identify, explain and apply the important ideas behind start menu and pinned apps.



Five important ideas

Desktop: is a main computer work area.

Icon: is a small picture representing an item.

Taskbar: helps find open apps and tools.

Wallpaper: is the desktop background picture.

Start menu: offers a way to open programs.

Our Ghanaian story

When the computer starts, Yaw had a question about start menu and pinned apps. The group discovered that desktop is a main computer work area, while icon is a small picture representing an item. They drew an example and explained it to their classmates.

Try it!

With an adult supervising, practise desktop and icon using a computer if one is available. Without a computer, draw the screen and explain each step to a partner.

Create and share

Make a colourful, labelled poster that teaches start menu and pinned apps. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.2.1.2

Lesson 9: Play & Challenge

PLAY — Choose one best answer for each question.

1. Start Menu and Pinned Apps — Choose the correct name for something that is a main computer work area.

A. Desktop B. Taskbar C. Wallpaper D. Icon

2. Start Menu and Pinned Apps — Your friend says, "This is a small picture representing an item." What is the friend describing?

A. Desktop B. Icon C. Wallpaper D. Taskbar

3. Start Menu and Pinned Apps — Which idea from start menu and pinned apps matches the clue "helps find open apps and tools"?

A. Icon B. Desktop C. Wallpaper D. Taskbar

4. Start Menu and Pinned Apps — The teacher asks for an example that is the desktop background picture. Which answer fits?

A. Desktop B. Start menu C. Icon D. Wallpaper

5. Start Menu and Pinned Apps — Can you identify the term described here: offers a way to open programs?

A. Icon B. Wallpaper C. Start menu D. Desktop

CHALLENGE — Think and apply.

1. Start Menu and Pinned Apps — A group makes a display on start menu and pinned apps. Their first clue is "is a small picture representing an item". Which label is correct?

A. Wallpaper B. Taskbar C. Desktop D. Icon

2. Start Menu and Pinned Apps — At home, Esi remembers a class rule or concept that is the desktop background picture. What was it called?

A. Start menu B. Wallpaper C. Icon D. Desktop

3. Start Menu and Pinned Apps — A classmate confuses several ideas about start menu and pinned apps. Which one actually is a small picture representing an item?

A. Desktop B. Icon C. Wallpaper D. Taskbar

4. Start Menu and Pinned Apps — During a school ICT lesson, Esi wants to show something that is a main computer work area. Which idea should be labelled?

A. Desktop B. Taskbar C. Icon D. Wallpaper

5. Start Menu and Pinned Apps — Which answer would fit on a poster beside the words "is a main computer work area"?

A. Icon B. Taskbar C. Desktop D. Wallpaper

6. Start Menu and Pinned Apps — At the community learning centre, someone describes a tool or idea that offers a way to open programs. Which one is it?

A. Wallpaper B. Start menu C. Desktop D. Icon

7. Start Menu and Pinned Apps — Imagine helping a younger pupil understand something that helps find open apps and tools. Which term would you teach?

A. Taskbar B. Wallpaper C. Desktop D. Icon

8. Start Menu and Pinned Apps — Your class is solving a practical problem. One useful thing is the desktop background picture. What is its name?

A. Desktop B. Start menu C. Wallpaper D. Icon

9. Start Menu and Pinned Apps — Which concept matches the explanation "offers a way to open programs" when you review start menu and pinned apps?

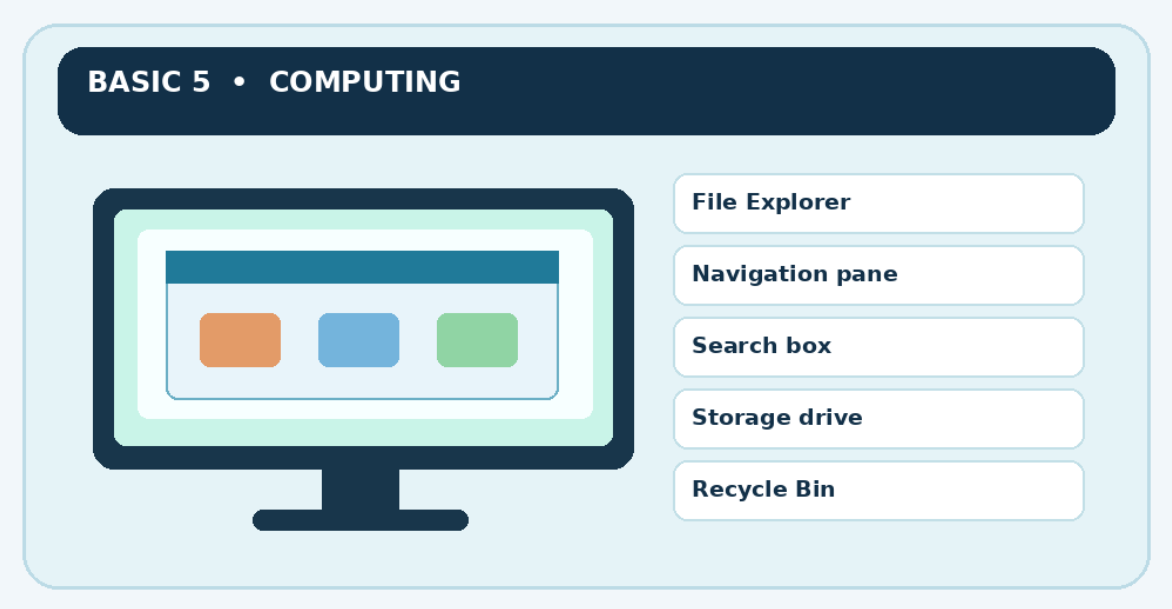
A. Wallpaper B. Start menu C. Desktop D. Icon

10. Start Menu and Pinned Apps — For a short demonstration, the teacher needs an example that helps find open apps and tools. Which choice would you select?

A. Desktop B. Icon C. Wallpaper D. Taskbar

Lesson 10: Desktop Shortcuts

We will learn to: Identify, explain and apply the important ideas behind desktop shortcuts.



Five important ideas

File Explorer: helps browse computer files and folders.

Navigation pane: lists useful locations.

Search box: helps find a named item.

Storage drive: holds saved digital information.

Recycle Bin: temporarily holds many deleted files.

Our Ghanaian story

While searching for classwork, Abena had a question about desktop shortcuts. The group discovered that file explorer helps browse computer files and folders, while navigation pane lists useful locations. They drew an example and explained it to their classmates.

Try it!

With an adult supervising, practise file explorer and navigation pane using a computer if one is available. Without a computer, draw the screen and explain each step to a partner.

Create and share

Make a colourful, labelled poster that teaches desktop shortcuts. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.2.1.4

Lesson 10: Play & Challenge

PLAY — Choose one best answer for each question.

1. Desktop Shortcuts — Choose the correct name for something that helps browse computer files and folders.

A. Storage drive B. Recycle Bin C. File Explorer D. Navigation pane

2. Desktop Shortcuts — Your friend says, "This lists useful locations." What is the friend describing?

A. File Explorer B. Recycle Bin C. Navigation pane D. Storage drive

3. Desktop Shortcuts — Which idea from desktop shortcuts matches the clue "helps find a named item"?

A. Recycle Bin B. Search box C. Navigation pane D. Storage drive

4. Desktop Shortcuts — The teacher asks for an example that holds saved digital information. Which answer fits?

A. File Explorer B. Search box C. Storage drive D. Navigation pane

5. Desktop Shortcuts — Can you identify the term described here: temporarily holds many deleted files?

A. Recycle Bin B. Search box C. Navigation pane D. Storage drive

CHALLENGE — Think and apply.

1. Desktop Shortcuts — A group makes a display on desktop shortcuts. Their first clue is "lists useful locations". Which label is correct?

A. Recycle Bin B. Navigation pane C. Storage drive D. File Explorer

2. Desktop Shortcuts — Imagine helping a younger pupil understand something that helps find a named item. Which term would you teach?

A. Search box B. Storage drive C. Navigation pane D. Recycle Bin

3. Desktop Shortcuts — At home, Ama remembers a class rule or concept that holds saved digital information. What was it called?

A. File Explorer B. Navigation pane C. Storage drive D. Search box

4. Desktop Shortcuts — Which answer would fit on a poster beside the words "helps browse computer files and folders"?

A. Recycle Bin B. File Explorer C. Navigation pane D. Storage drive

5. Desktop Shortcuts — At the community learning centre, someone describes a tool or idea that temporarily holds many deleted files. Which one is it?

A. Navigation pane B. Storage drive C. Search box D. Recycle Bin

6. Desktop Shortcuts — Your class is solving a practical problem. One useful thing holds saved digital information. What is its name?

A. Search box B. File Explorer C. Storage drive D. Navigation pane

7. Desktop Shortcuts — A classmate confuses several ideas about desktop shortcuts. Which one actually lists useful locations?

A. Navigation pane B. File Explorer C. Storage drive D. Recycle Bin

8. Desktop Shortcuts — During a school ICT lesson, Ama wants to show something that helps browse computer files and folders. Which idea should be labelled?

A. Navigation pane B. Recycle Bin C. Storage drive D. File Explorer

9. Desktop Shortcuts — Which concept matches the explanation "temporarily holds many deleted files" when you review desktop shortcuts?

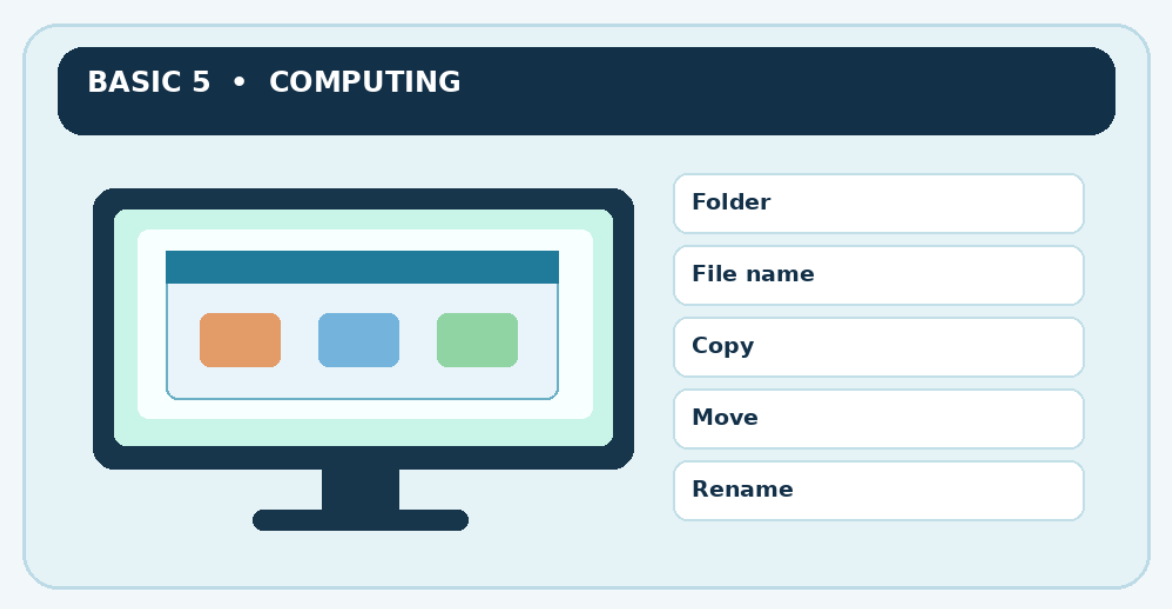
A. Recycle Bin B. Navigation pane C. Search box D. Storage drive

10. Desktop Shortcuts — For a short demonstration, the teacher needs an example that helps find a named item. Which choice would you select?

A. Storage drive B. Navigation pane C. Recycle Bin D. Search box

Lesson 11: Move and Copy Files

We will learn to: Identify, explain and apply the important ideas behind move and copy files.



Five important ideas

Folder: organises related digital files.

File name: helps identify a saved item.

Copy: makes a second version of an item.

Move: changes where an item is stored.

Rename: changes the label without changing the file content.

Our Ghanaian story

While organising class projects, Kofi had a question about move and copy files. The group discovered that folder organises related digital files, while file name helps identify a saved item. They drew an example and explained it to their classmates.

Try it!

With an adult supervising, practise folder and file name using a computer if one is available. Without a computer, draw the screen and explain each step to a partner.

Create and share

Make a colourful, labelled poster that teaches move and copy files. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.2.1.5

Lesson 11: Play & Challenge

PLAY — Choose one best answer for each question.

1. Move and Copy Files — Choose the correct name for something that organises related digital files.

A. File name B. Move C. Folder D. Copy

2. Move and Copy Files — Your friend says, "This helps identify a saved item." What is the friend describing?

A. Rename B. Move C. Copy D. File name

3. Move and Copy Files — Which idea from move and copy files matches the clue "makes a second version of an item"?

A. Move B. Folder C. Copy D. Rename

4. Move and Copy Files — The teacher asks for an example that changes where an item is stored. Which answer fits?

A. File name B. Move C. Copy D. Folder

5. Move and Copy Files — Can you identify the term described here: changes the label without changing the file content?

A. File name B. Copy C. Rename D. Folder

CHALLENGE — Think and apply.

1. Move and Copy Files — At the community learning centre, someone describes a tool or idea that changes the label without changing the file content. Which one is it?

A. File name B. Folder C. Copy D. Rename

2. Move and Copy Files — Your class is solving a practical problem. One useful thing changes where an item is stored. What is its name?

A. Copy B. File name C. Move D. Folder

3. Move and Copy Files — A classmate confuses several ideas about move and copy files. Which one actually helps identify a saved item?

A. File name B. Move C. Copy D. Rename

4. Move and Copy Files — Imagine helping a younger pupil understand something that makes a second version of an item. Which term would you teach?

A. Folder B. Rename C. Copy D. Move

5. Move and Copy Files — Which concept matches the explanation "changes the label without changing the file content" when you review move and copy files?

A. Folder B. Copy C. Rename D. File name

6. Move and Copy Files — A group makes a display on move and copy files. Their first clue is "helps identify a saved item". Which label is correct?

A. Copy B. Rename C. Move D. File name

7. Move and Copy Files — Which answer would fit on a poster beside the words "organises related digital files"?

A. Move B. File name C. Copy D. Folder

8. Move and Copy Files — At home, Yaw remembers a class rule or concept that changes where an item is stored. What was it called?

A. Folder B. Move C. File name D. Copy

9. Move and Copy Files — For a short demonstration, the teacher needs an example that makes a second version of an item. Which choice would you select?

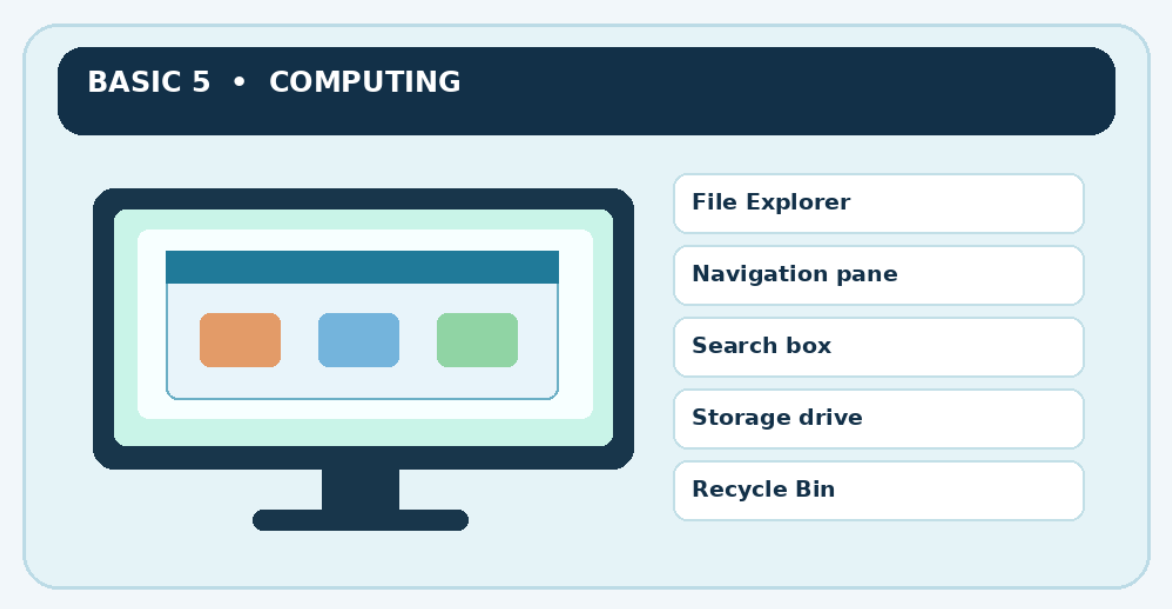
A. Rename B. Copy C. Folder D. Move

10. Move and Copy Files — During a school ICT lesson, Yaw wants to show something that organises related digital files. Which idea should be labelled?

A. Folder B. Move C. File name D. Copy

Lesson 12: Use File Explorer

We will learn to: Identify, explain and apply the important ideas behind use file explorer.



Five important ideas

File Explorer: helps browse computer files and folders.

Navigation pane: lists useful locations.

Search box: helps find a named item.

Storage drive: holds saved digital information.

Recycle Bin: temporarily holds many deleted files.

Our Ghanaian story

While searching for classwork, Ama had a question about use file explorer. The group discovered that file explorer helps browse computer files and folders, while navigation pane lists useful locations. They drew an example and explained it to their classmates.

Try it!

With an adult supervising, practise file explorer and navigation pane using a computer if one is available. Without a computer, draw the screen and explain each step to a partner.

Create and share

Make a colourful, labelled poster that teaches use file explorer. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.2.1.6

Lesson 12: Play & Challenge

PLAY — Choose one best answer for each question.

1. Use File Explorer — Choose the correct name for something that helps browse computer files and folders.

A. File Explorer B. Search box C. Storage drive D. Recycle Bin

2. Use File Explorer — Your friend says, "This lists useful locations." What is the friend describing?

A. Recycle Bin B. Navigation pane C. Search box D. File Explorer

3. Use File Explorer — Which idea from use file explorer matches the clue "helps find a named item"?

A. Search box B. File Explorer C. Navigation pane D. Storage drive

4. Use File Explorer — The teacher asks for an example that holds saved digital information. Which answer fits?

A. Navigation pane B. Storage drive C. Recycle Bin D. Search box

5. Use File Explorer — Can you identify the term described here: temporarily holds many deleted files?

A. Recycle Bin B. Search box C. Navigation pane D. File Explorer

CHALLENGE — Think and apply.

1. Use File Explorer — Your class is solving a practical problem. One useful thing holds saved digital information. What is its name?

A. Search box B. Recycle Bin C. Navigation pane D. Storage drive

2. Use File Explorer — Which concept matches the explanation "temporarily holds many deleted files" when you review use file explorer?

A. Search box B. Recycle Bin C. Navigation pane D. File Explorer

3. Use File Explorer — At home, Kofi remembers a class rule or concept that holds saved digital information. What was it called?

A. Navigation pane B. Recycle Bin C. Search box D. Storage drive

4. Use File Explorer — Imagine helping a younger pupil understand something that helps find a named item. Which term would you teach?

A. File Explorer B. Search box C. Navigation pane D. Storage drive

5. Use File Explorer — Which answer would fit on a poster beside the words "helps browse computer files and folders"?

A. Recycle Bin B. Storage drive C. File Explorer D. Search box

6. Use File Explorer — A classmate confuses several ideas about use file explorer. Which one actually lists useful locations?

A. File Explorer B. Navigation pane C. Recycle Bin D. Search box

7. Use File Explorer — At the community learning centre, someone describes a tool or idea that temporarily holds many deleted files. Which one is it?

A. Recycle Bin B. Search box C. File Explorer D. Navigation pane

8. Use File Explorer — A group makes a display on use file explorer. Their first clue is "lists useful locations". Which label is correct?

A. File Explorer B. Navigation pane C. Recycle Bin D. Search box

9. Use File Explorer — For a short demonstration, the teacher needs an example that helps find a named item. Which choice would you select?

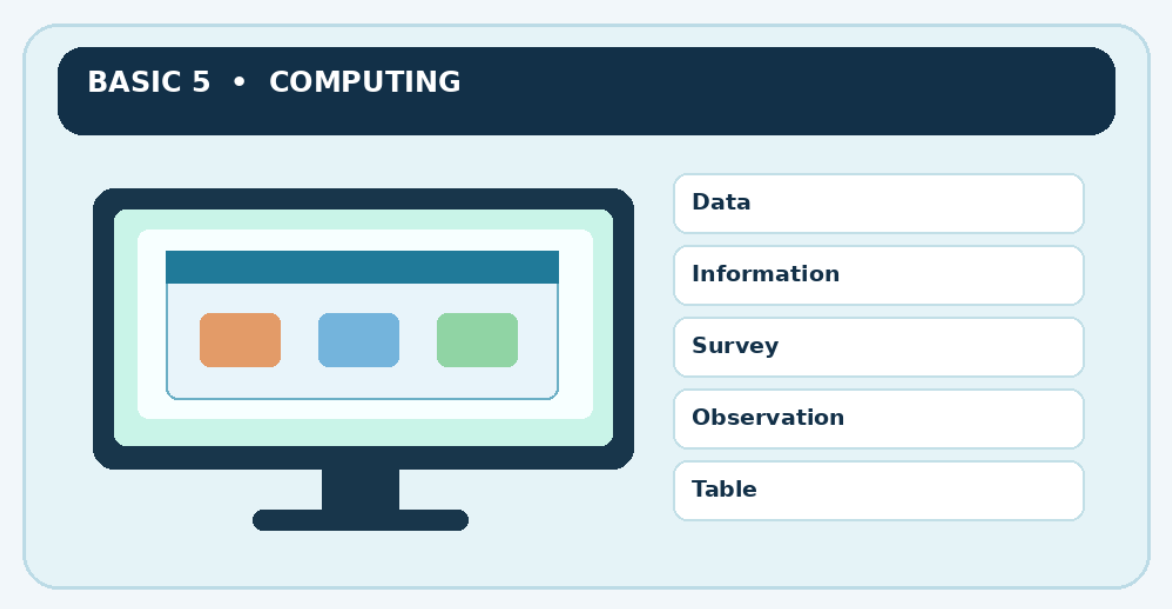
A. File Explorer B. Storage drive C. Search box D. Navigation pane

10. Use File Explorer — During a school ICT lesson, Kofi wants to show something that helps browse computer files and folders. Which idea should be labelled?

A. Recycle Bin B. Storage drive C. Search box D. File Explorer

Lesson 13: Data From Experiments

We will learn to: Identify, explain and apply the important ideas behind data from experiments.



Five important ideas

Data: raw facts gathered for a purpose.

Information: data organised to make sense.

Survey: asks people selected questions.

Observation: records what someone notices.

Table: arranges facts into rows and columns.

Our Ghanaian story

During a survey of the school, Kojo had a question about data from experiments. The group discovered that data raw facts gathered for a purpose, while information data organised to make sense. They drew an example and explained it to their classmates.

Try it!

Collect a few harmless facts about classroom materials, NOT personal details. Make a paper chart with the terms Data, Information and Survey; explain what each means.

Create and share

Make a colourful, labelled poster that teaches data from experiments. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.3.1.1

Lesson 13: Play & Challenge

PLAY — Choose one best answer for each question.

1. Data From Experiments — Choose the correct name for something that raw facts gathered for a purpose.

A. Survey B. Observation C. Information D. Data

2. Data From Experiments — Your friend says, "This data organised to make sense." What is the friend describing?

A. Observation B. Information C. Data D. Survey

3. Data From Experiments — Which idea from data from experiments matches the clue "asks people selected questions"?

A. Information B. Observation C. Survey D. Table

4. Data From Experiments — The teacher asks for an example that records what someone notices. Which answer fits?

A. Survey B. Observation C. Table D. Data

5. Data From Experiments — Can you identify the term described here: arranges facts into rows and columns?

A. Table B. Data C. Information D. Observation

CHALLENGE — Think and apply.

1. Data From Experiments — For a short demonstration, the teacher needs an example that asks people selected questions. Which choice would you select?

A. Information B. Table C. Observation D. Survey

2. Data From Experiments — Imagine helping a younger pupil understand something that asks people selected questions. Which term would you teach?

A. Observation B. Survey C. Table D. Information

3. Data From Experiments — Which concept matches the explanation "arranges facts into rows and columns" when you review data from experiments?

A. Information B. Observation C. Data D. Table

4. Data From Experiments — Your class is solving a practical problem. One useful thing records what someone notices. What is its name?

A. Data B. Table C. Observation D. Survey

5. Data From Experiments — During a school ICT lesson, Akosua wants to show something that raw facts gathered for a purpose. Which idea should be labelled?

A. Data B. Observation C. Information D. Survey

6. Data From Experiments — At home, Akosua remembers a class rule or concept that records what someone notices. What was it called?

A. Survey B. Table C. Data D. Observation

7. Data From Experiments — Which answer would fit on a poster beside the words "raw facts gathered for a purpose"?

A. Observation B. Information C. Survey D. Data

8. Data From Experiments — A group makes a display on data from experiments. Their first clue is "data organised to make sense". Which label is correct?

A. Observation B. Survey C. Data D. Information

9. Data From Experiments — At the community learning centre, someone describes a tool or idea that arranges facts into rows and columns. Which one is it?

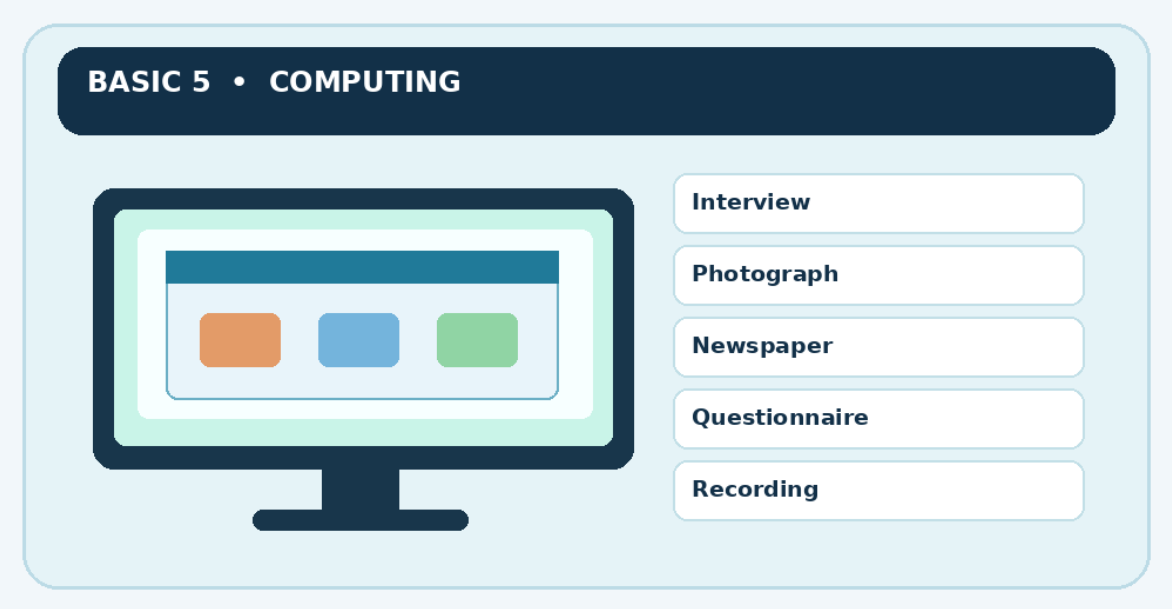
A. Data B. Information C. Observation D. Table

10. Data From Experiments — A classmate confuses several ideas about data from experiments. Which one actually data organised to make sense?

A. Survey B. Observation C. Data D. Information

Lesson 14: Primary Sources of Data

We will learn to: Identify, explain and apply the important ideas behind primary sources of data.



Five important ideas

Interview: collects answers by talking to people.

Photograph: records a visual moment.

Newspaper: reports news and events.

Questionnaire: collects written responses.

Recording: stores captured sound or video.

Our Ghanaian story

When preparing a report about the neighbourhood, Esi had a question about primary sources of data. The group discovered that interview collects answers by talking to people, while photograph records a visual moment. They drew an example and explained it to their classmates.

Try it!

Collect a few harmless facts about classroom materials, NOT personal details. Make a paper chart with the terms Interview, Photograph and Newspaper; explain what each means.

Create and share

Make a colourful, labelled poster that teaches primary sources of data. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.3.1.2

Lesson 14: Play & Challenge

PLAY — Choose one best answer for each question.

1. Primary Sources of Data — Choose the correct name for something that collects answers by talking to people.

A. Interview B. Photograph C. Recording D. Questionnaire

2. Primary Sources of Data — Your friend says, "This records a visual moment." What is the friend describing?

A. Photograph B. Recording C. Questionnaire D. Newspaper

3. Primary Sources of Data — Which idea from primary sources of data matches the clue "reports news and events"?

A. Photograph B. Newspaper C. Interview D. Recording

4. Primary Sources of Data — The teacher asks for an example that collects written responses. Which answer fits?

A. Interview B. Questionnaire C. Newspaper D. Photograph

5. Primary Sources of Data — Can you identify the term described here: stores captured sound or video?

A. Questionnaire B. Interview C. Recording D. Newspaper

CHALLENGE — Think and apply.

1. Primary Sources of Data — At home, Esi remembers a class rule or concept that collects written responses. What was it called?

A. Interview B. Photograph C. Questionnaire D. Newspaper

2. Primary Sources of Data — A group makes a display on primary sources of data. Their first clue is "records a visual moment". Which label is correct?

A. Photograph B. Newspaper C. Recording D. Questionnaire

3. Primary Sources of Data — For a short demonstration, the teacher needs an example that reports news and events. Which choice would you select?

A. Photograph B. Newspaper C. Recording D. Interview

4. Primary Sources of Data — Which concept matches the explanation "stores captured sound or video" when you review primary sources of data?

A. Newspaper B. Questionnaire C. Recording D. Interview

5. Primary Sources of Data — Which answer would fit on a poster beside the words "collects answers by talking to people"?

A. Recording B. Questionnaire C. Interview D. Photograph

6. Primary Sources of Data — At the community learning centre, someone describes a tool or idea that stores captured sound or video. Which one is it?

A. Interview B. Recording C. Questionnaire D. Newspaper

7. Primary Sources of Data — A classmate confuses several ideas about primary sources of data. Which one actually records a visual moment?

A. Recording B. Photograph C. Newspaper D. Questionnaire

8. Primary Sources of Data — During a school ICT lesson, Esi wants to show something that collects answers by talking to people. Which idea should be labelled?

A. Questionnaire B. Photograph C. Interview D. Recording

9. Primary Sources of Data — Your class is solving a practical problem. One useful thing collects written responses. What is its name?

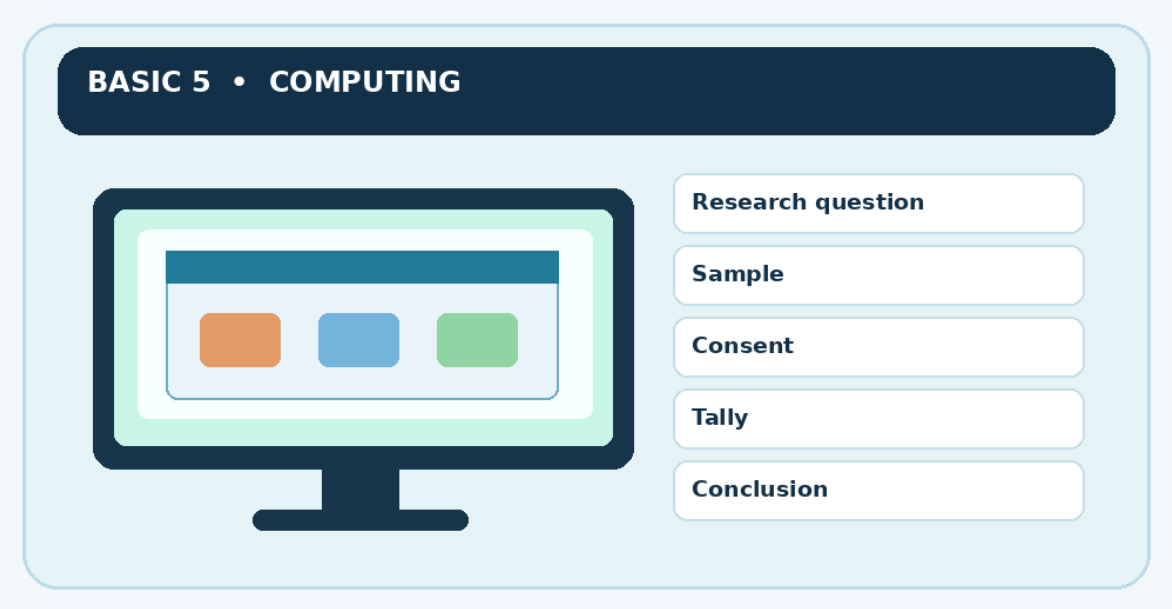
A. Questionnaire B. Interview C. Photograph D. Newspaper

10. Primary Sources of Data — Imagine helping a younger pupil understand something that reports news and events. Which term would you teach?

A. Interview B. Newspaper C. Recording D. Photograph

Lesson 15: Collect a Data Sample

We will learn to: Identify, explain and apply the important ideas behind collect a data sample.



Five important ideas

Research question: states what we want to discover.

Sample: is a smaller selected group.

Consent: means someone agrees to participate.

Tally: records counts using marks.

Conclusion: explains what the collected results suggest.

Our Ghanaian story

When interviewing classmates about favourite fruits, Yaw had a question about collect a data sample. The group discovered that research question states what we want to discover, while sample is a smaller selected group. They drew an example and explained it to their classmates.

Try it!

Collect a few harmless facts about classroom materials, NOT personal details. Make a paper chart with the terms Research question, Sample and Consent; explain what each means.

Create and share

Make a colourful, labelled poster that teaches collect a data sample. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.3.1.8

Lesson 15: Play & Challenge

PLAY — Choose one best answer for each question.

1. Collect a Data Sample — Choose the correct name for something that states what we want to discover.

A. Consent B. Conclusion C. Research question D. Sample

2. Collect a Data Sample — Your friend says, "This is a smaller selected group." What is the friend describing?

A. Sample B. Research question C. Tally D. Conclusion

3. Collect a Data Sample — Which idea from collect a data sample matches the clue "means someone agrees to participate"?

A. Research question B. Tally C. Sample D. Consent

4. Collect a Data Sample — The teacher asks for an example that records counts using marks. Which answer fits?

A. Consent B. Tally C. Conclusion D. Research question

5. Collect a Data Sample — Can you identify the term described here: explains what the collected results suggest?

A. Research question B. Sample C. Conclusion D. Consent

CHALLENGE — Think and apply.

1. Collect a Data Sample — At the community learning centre, someone describes a tool or idea that explains what the collected results suggest. Which one is it?

A. Research question B. Consent C. Sample D. Conclusion

2. Collect a Data Sample — During a school ICT lesson, Ama wants to show something that states what we want to discover. Which idea should be labelled?

A. Consent B. Research question C. Sample D. Conclusion

3. Collect a Data Sample — A classmate confuses several ideas about collect a data sample. Which one actually is a smaller selected group?

A. Research question B. Sample C. Conclusion D. Tally

4. Collect a Data Sample — Which concept matches the explanation "explains what the collected results suggest" when you review collect a data sample?

A. Conclusion B. Consent C. Research question D. Sample

5. Collect a Data Sample — Imagine helping a younger pupil understand something that means someone agrees to participate. Which term would you teach?

A. Consent B. Sample C. Research question D. Tally

6. Collect a Data Sample — At home, Ama remembers a class rule or concept that records counts using marks. What was it called?

A. Conclusion B. Tally C. Consent D. Research question

7. Collect a Data Sample — Your class is solving a practical problem. One useful thing records counts using marks. What is its name?

A. Tally B. Conclusion C. Research question D. Consent

8. Collect a Data Sample — For a short demonstration, the teacher needs an example that means someone agrees to participate. Which choice would you select?

A. Sample B. Tally C. Research question D. Consent

9. Collect a Data Sample — A group makes a display on collect a data sample. Their first clue is "is a smaller selected group". Which label is correct?

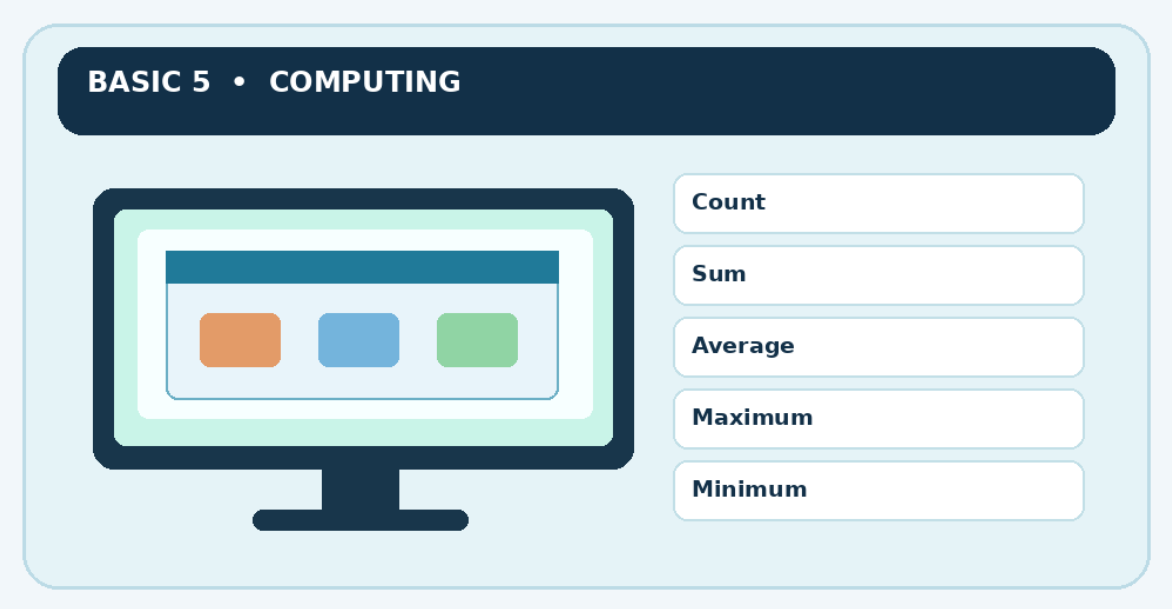
A. Tally B. Research question C. Sample D. Conclusion

10. Collect a Data Sample — Which answer would fit on a poster beside the words "states what we want to discover"?

A. Conclusion B. Sample C. Research question D. Consent

Lesson 16: Analyse Charts and Tables

We will learn to: Identify, explain and apply the important ideas behind analyse charts and tables.



Five important ideas

Count: tells how many items are present.

Sum: adds numbers together.

Average: shares a total evenly across the number of values.

Maximum: is the largest value.

Minimum: is the smallest value.

Our Ghanaian story

While calculating scores for sports day, Abena had a question about analyse charts and tables. The group discovered that count tells how many items are present, while sum adds numbers together. They drew an example and explained it to their classmates.

Try it!

Collect a few harmless facts about classroom materials, NOT personal details. Make a paper chart with the terms Count, Sum and Average; explain what each means.

Create and share

Make a colourful, labelled poster that teaches analyse charts and tables. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.3.1.10

Lesson 16: Play & Challenge

PLAY — Choose one best answer for each question.

1. Analyse Charts and Tables — Choose the correct name for something that tells how many items are present.

A. Count B. Sum C. Maximum D. Average

2. Analyse Charts and Tables — Your friend says, "This adds numbers together." What is the friend describing?

A. Minimum B. Average C. Count D. Sum

3. Analyse Charts and Tables — Which idea from analyse charts and tables matches the clue "shares a total evenly across the number of values"?

A. Minimum B. Average C. Count D. Sum

4. Analyse Charts and Tables — The teacher asks for an example that is the largest value. Which answer fits?

A. Average B. Sum C. Maximum D. Minimum

5. Analyse Charts and Tables — Can you identify the term described here: is the smallest value?

A. Average B. Minimum C. Maximum D. Count

CHALLENGE — Think and apply.

1. Analyse Charts and Tables — At the community learning centre, someone describes a tool or idea that is the smallest value. Which one is it?

A. Count B. Maximum C. Average D. Minimum

2. Analyse Charts and Tables — Imagine helping a younger pupil understand something that shares a total evenly across the number of values. Which term would you teach?

A. Average B. Minimum C. Count D. Sum

3. Analyse Charts and Tables — At home, Yaw remembers a class rule or concept that is the largest value. What was it called?

A. Average B. Sum C. Minimum D. Maximum

4. Analyse Charts and Tables — During a school ICT lesson, Yaw wants to show something that tells how many items are present. Which idea should be labelled?

A. Maximum B. Average C. Count D. Sum

5. Analyse Charts and Tables — Your class is solving a practical problem. One useful thing is the largest value. What is its name?

A. Maximum B. Sum C. Minimum D. Average

6. Analyse Charts and Tables — Which answer would fit on a poster beside the words "tells how many items are present"?

A. Sum B. Count C. Average D. Maximum

7. Analyse Charts and Tables — A classmate confuses several ideas about analyse charts and tables. Which one actually adds numbers together?

A. Sum B. Average C. Count D. Minimum

8. Analyse Charts and Tables — A group makes a display on analyse charts and tables. Their first clue is "adds numbers together". Which label is correct?

A. Minimum B. Sum C. Average D. Count

9. Analyse Charts and Tables — For a short demonstration, the teacher needs an example that shares a total evenly across the number of values. Which choice would you select?

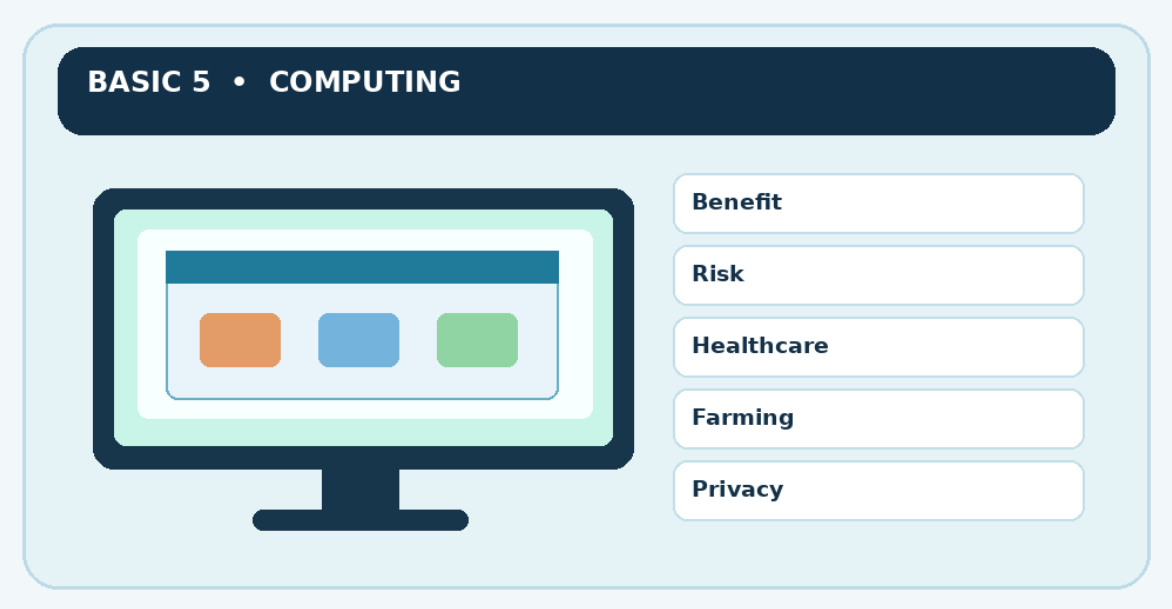
A. Sum B. Average C. Count D. Minimum

10. Analyse Charts and Tables — Which concept matches the explanation "is the smallest value" when you review analyse charts and tables?

A. Count B. Average C. Minimum D. Maximum

Lesson 17: How Technology Affects Communities

We will learn to: Identify, explain and apply the important ideas behind how technology affects communities.



Five important ideas

Benefit: is a helpful effect of technology.

Risk: is a possible harmful result.

Healthcare: can use technology to support patients.

Farming: can use sensors and weather data.

Privacy: means controlling access to personal information.

Our Ghanaian story

While discussing changes in a Ghanaian community, Kofi had a question about how technology affects communities. The group discovered that benefit is a helpful effect of technology, while risk is a possible harmful result. They drew an example and explained it to their classmates.

Try it!

Create a class safety demonstration. Show a good example of benefit and risk and explain why the good practice matters.

Create and share

Make a colourful, labelled poster that teaches how technology affects communities. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.4.1.1

Lesson 17: Play & Challenge

PLAY — Choose one best answer for each question.

1. How Technology Affects Communities — Choose the correct name for something that is a helpful effect of technology.

A. Farming B. Privacy C. Benefit D. Risk

2. How Technology Affects Communities — Your friend says, "This is a possible harmful result." What is the friend describing?

A. Benefit B. Farming C. Risk D. Privacy

3. How Technology Affects Communities — Which idea from how technology affects communities matches the clue "can use technology to support patients"?

A. Risk B. Healthcare C. Benefit D. Farming

4. How Technology Affects Communities — The teacher asks for an example that can use sensors and weather data. Which answer fits?

A. Benefit B. Healthcare C. Risk D. Farming

5. How Technology Affects Communities — Can you identify the term described here: means controlling access to personal information?

A. Risk B. Benefit C. Privacy D. Healthcare

CHALLENGE — Think and apply.

1. How Technology Affects Communities — Your class is solving a practical problem. One useful thing can use sensors and weather data. What is its name?

A. Healthcare B. Benefit C. Farming D. Risk

2. How Technology Affects Communities — For a short demonstration, the teacher needs an example that can use technology to support patients. Which choice would you select?

A. Farming B. Benefit C. Healthcare D. Risk

3. How Technology Affects Communities — A classmate confuses several ideas about how technology affects communities. Which one actually is a possible harmful result?

A. Privacy B. Farming C. Risk D. Benefit

4. How Technology Affects Communities — At the community learning centre, someone describes a tool or idea that means controlling access to personal information. Which one is it?

A. Benefit B. Privacy C. Risk D. Healthcare

5. How Technology Affects Communities — Which answer would fit on a poster beside the words "is a helpful effect of technology"?

A. Farming B. Risk C. Privacy D. Benefit

6. How Technology Affects Communities — At home, Kofi remembers a class rule or concept that can use sensors and weather data. What was it called?

A. Risk B. Healthcare C. Benefit D. Farming

7. How Technology Affects Communities — Imagine helping a younger pupil understand something that can use technology to support patients. Which term would you teach?

A. Benefit B. Farming C. Risk D. Healthcare

8. How Technology Affects Communities — A group makes a display on how technology affects communities. Their first clue is "is a possible harmful result". Which label is correct?

A. Risk B. Benefit C. Privacy D. Farming

9. How Technology Affects Communities — Which concept matches the explanation "means controlling access to personal information" when you review how technology affects communities?

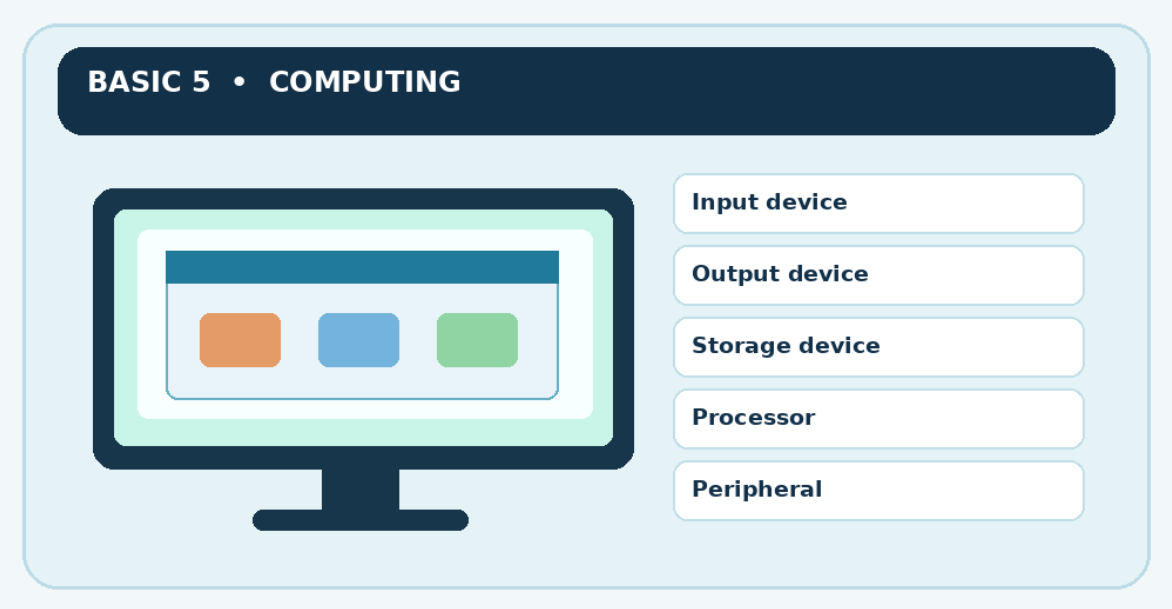
A. Healthcare B. Benefit C. Risk D. Privacy

10. How Technology Affects Communities — During a school ICT lesson, Kofi wants to show something that is a helpful effect of technology. Which idea should be labelled?

A. Benefit B. Farming C. Privacy D. Risk

Lesson 18: Hardware, Software and Networks

We will learn to: Identify, explain and apply the important ideas behind hardware, software and networks.



Five important ideas

Input device: sends information into a computer.

Output device: shows or plays results.

Storage device: keeps digital files.

Processor: helps follow computing instructions.

Peripheral: is an extra device connected to a computer.

Our Ghanaian story

During a tour of the computer room, Ama had a question about hardware, software and networks. The group discovered that input device sends information into a computer, while output device shows or plays results. They drew an example and explained it to their classmates.

Try it!

In a small group, draw or safely examine pictures of input device, output device and storage device. Explain one real-life use for each in Ghana.

Create and share

Make a colourful, labelled poster that teaches hardware, software and networks. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.1.4.1.2

Lesson 18: Play & Challenge

PLAY — Choose one best answer for each question.

1. Hardware, Software and Networks — Choose the correct name for something that sends information into a computer.

A. Input device B. Processor C. Storage device D. Output device

2. Hardware, Software and Networks — Your friend says, "This shows or plays results." What is the friend describing?

A. Storage device B. Output device C. Processor D. Peripheral

3. Hardware, Software and Networks — Which idea from hardware, software and networks matches the clue "keeps digital files"?

A. Output device B. Storage device C. Input device D. Processor

4. Hardware, Software and Networks — The teacher asks for an example that helps follow computing instructions. Which answer fits?

A. Processor B. Storage device C. Peripheral D. Output device

5. Hardware, Software and Networks — Can you identify the term described here: is an extra device connected to a computer?

A. Peripheral B. Processor C. Input device D. Storage device

CHALLENGE — Think and apply.

1. Hardware, Software and Networks — At the community learning centre, someone describes a tool or idea that is an extra device connected to a computer. Which one is it?

A. Storage device B. Peripheral C. Input device D. Processor

2. Hardware, Software and Networks — Imagine helping a younger pupil understand something that keeps digital files. Which term would you teach?

A. Processor B. Storage device C. Output device D. Input device

3. Hardware, Software and Networks — Your class is solving a practical problem. One useful thing helps follow computing instructions. What is its name?

A. Output device B. Peripheral C. Processor D. Storage device

4. Hardware, Software and Networks — For a short demonstration, the teacher needs an example that keeps digital files. Which choice would you select?

A. Input device B. Storage device C. Processor D. Output device

5. Hardware, Software and Networks — A classmate confuses several ideas about hardware, software and networks. Which one actually shows or plays results?

A. Output device B. Processor C. Peripheral D. Storage device

6. Hardware, Software and Networks — During a school ICT lesson, Akosua wants to show something that sends information into a computer. Which idea should be labelled?

A. Input device B. Processor C. Storage device D. Output device

7. Hardware, Software and Networks — Which answer would fit on a poster beside the words "sends information into a computer"?

A. Output device B. Input device C. Processor D. Storage device

8. Hardware, Software and Networks — A group makes a display on hardware, software and networks. Their first clue is "shows or plays results". Which label is correct?

A. Output device B. Storage device C. Processor D. Peripheral

9. Hardware, Software and Networks — At home, Akosua remembers a class rule or concept that helps follow computing instructions. What was it called?

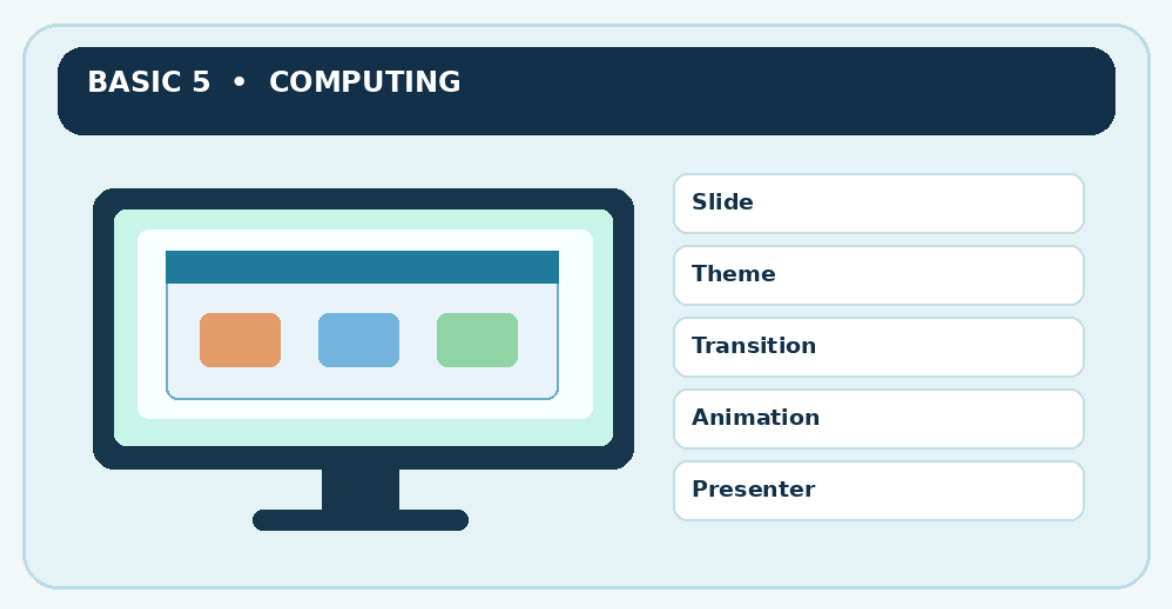
A. Peripheral B. Processor C. Output device D. Storage device

10. Hardware, Software and Networks — Which concept matches the explanation "is an extra device connected to a computer" when you review hardware, software and networks?

A. Peripheral B. Storage device C. Processor D. Input device

Lesson 19: PowerPoint Design and Animation

We will learn to: Identify, explain and apply the important ideas behind powerpoint design and animation.



Five important ideas

Slide: is one screen of a presentation.

Theme: gives slides a coordinated design.

Transition: controls how one slide changes to another.

Animation: makes an object appear or move.

Presenter: speaks clearly while showing the slides.

Our Ghanaian story

While preparing a school assembly talk, Kojo had a question about powerpoint design and animation. The group discovered that slide is one screen of a presentation, while theme gives slides a coordinated design. They drew an example and explained it to their classmates.

Try it!

Make a one-page paper example about your school or a Ghanaian festival. Include slide, theme and transition; present it clearly to a partner.

Create and share

Make a colourful, labelled poster that teaches powerpoint design and animation. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.2.1.1.1

Lesson 19: Play & Challenge

PLAY — Choose one best answer for each question.

1. PowerPoint Design and Animation — Choose the correct name for something that is one screen of a presentation.

A. Presenter B. Animation C. Slide D. Theme

2. PowerPoint Design and Animation — Your friend says, "This gives slides a coordinated design." What is the friend describing?

A. Slide B. Presenter C. Transition D. Theme

3. PowerPoint Design and Animation — Which idea from powerpoint design and animation matches the clue "controls how one slide changes to another"?

A. Animation B. Presenter C. Transition D. Slide

4. PowerPoint Design and Animation — The teacher asks for an example that makes an object appear or move. Which answer fits?

A. Presenter B. Theme C. Transition D. Animation

5. PowerPoint Design and Animation — Can you identify the term described here: speaks clearly while showing the slides?

A. Transition B. Presenter C. Slide D. Theme

CHALLENGE — Think and apply.

1. PowerPoint Design and Animation — During a school ICT lesson, Esi wants to show something that is one screen of a presentation. Which idea should be labelled?

A. Animation B. Slide C. Presenter D. Theme

2. PowerPoint Design and Animation — A group makes a display on powerpoint design and animation. Their first clue is "gives slides a coordinated design". Which label is correct?

A. Slide B. Presenter C. Transition D. Theme

3. PowerPoint Design and Animation — Which concept matches the explanation "speaks clearly while showing the slides" when you review powerpoint design and animation?

A. Theme B. Transition C. Presenter D. Slide

4. PowerPoint Design and Animation — At home, Esi remembers a class rule or concept that makes an object appear or move. What was it called?

A. Theme B. Transition C. Presenter D. Animation

5. PowerPoint Design and Animation — For a short demonstration, the teacher needs an example that controls how one slide changes to another. Which choice would you select?

A. Animation B. Presenter C. Transition D. Slide

6. PowerPoint Design and Animation — At the community learning centre, someone describes a tool or idea that speaks clearly while showing the slides. Which one is it?

A. Presenter B. Transition C. Slide D. Theme

7. PowerPoint Design and Animation — Which answer would fit on a poster beside the words "is one screen of a presentation"?

A. Presenter B. Slide C. Theme D. Animation

8. PowerPoint Design and Animation — A classmate confuses several ideas about powerpoint design and animation. Which one actually gives slides a coordinated design?

A. Slide B. Transition C. Theme D. Presenter

9. PowerPoint Design and Animation — Your class is solving a practical problem. One useful thing makes an object appear or move. What is its name?

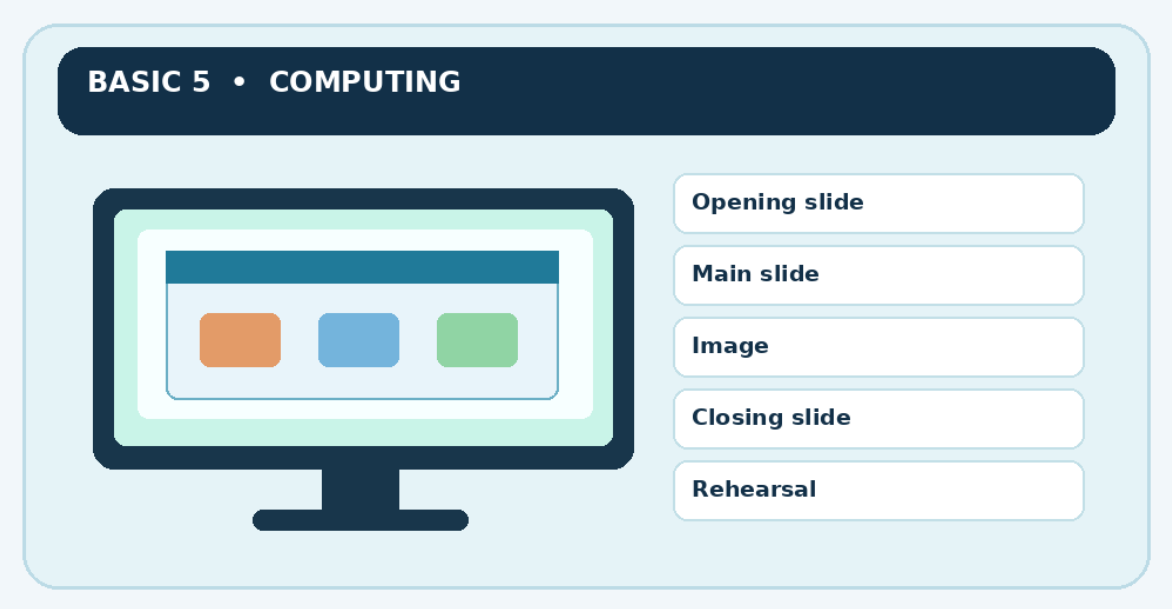
A. Transition B. Theme C. Presenter D. Animation

10. PowerPoint Design and Animation — Imagine helping a younger pupil understand something that controls how one slide changes to another. Which term would you teach?

A. Presenter B. Animation C. Transition D. Slide

Lesson 20: Present Five Slides

We will learn to: Identify, explain and apply the important ideas behind present five slides.



Five important ideas

Opening slide: introduces the topic.

Main slide: shows one key idea.

Image: helps communicate meaning.

Closing slide: sums up the message.

Rehearsal: helps a presenter speak with confidence.

Our Ghanaian story

Before presenting an environmental club project, Esi had a question about present five slides. The group discovered that opening slide introduces the topic, while main slide shows one key idea. They drew an example and explained it to their classmates.

Try it!

Make a one-page paper example about your school or a Ghanaian festival. Include opening slide, main slide and image; present it clearly to a partner.

Create and share

Make a colourful, labelled poster that teaches present five slides. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.2.1.1.2

Lesson 20: Play & Challenge

PLAY — Choose one best answer for each question.

1. Present Five Slides — Choose the correct name for something that introduces the topic.

A. Closing slide B. Opening slide C. Image D. Main slide

2. Present Five Slides — Your friend says, "This shows one key idea." What is the friend describing?

A. Image B. Rehearsal C. Main slide D. Closing slide

3. Present Five Slides — Which idea from present five slides matches the clue "helps communicate meaning"?

A. Main slide B. Closing slide C. Image D. Rehearsal

4. Present Five Slides — The teacher asks for an example that sums up the message. Which answer fits?

A. Closing slide B. Main slide C. Opening slide D. Rehearsal

5. Present Five Slides — Can you identify the term described here: helps a presenter speak with confidence?

A. Closing slide B. Main slide C. Image D. Rehearsal

CHALLENGE — Think and apply.

1. Present Five Slides — During a school ICT lesson, Ama wants to show something that introduces the topic. Which idea should be labelled?

A. Main slide B. Closing slide C. Opening slide D. Image

2. Present Five Slides — Your class is solving a practical problem. One useful thing sums up the message. What is its name?

A. Opening slide B. Main slide C. Closing slide D. Rehearsal

3. Present Five Slides — For a short demonstration, the teacher needs an example that helps communicate meaning. Which choice would you select?

A. Image B. Rehearsal C. Main slide D. Closing slide

4. Present Five Slides — Which concept matches the explanation "helps a presenter speak with confidence" when you review present five slides?

A. Main slide B. Image C. Rehearsal D. Closing slide

5. Present Five Slides — At the community learning centre, someone describes a tool or idea that helps a presenter speak with confidence. Which one is it?

A. Main slide B. Rehearsal C. Image D. Closing slide

6. Present Five Slides — Imagine helping a younger pupil understand something that helps communicate meaning. Which term would you teach?

A. Image B. Main slide C. Closing slide D. Rehearsal

7. Present Five Slides — A classmate confuses several ideas about present five slides. Which one actually shows one key idea?

A. Rehearsal B. Image C. Main slide D. Closing slide

8. Present Five Slides — Which answer would fit on a poster beside the words "introduces the topic"?

A. Opening slide B. Closing slide C. Main slide D. Image

9. Present Five Slides — A group makes a display on present five slides. Their first clue is "shows one key idea". Which label is correct?

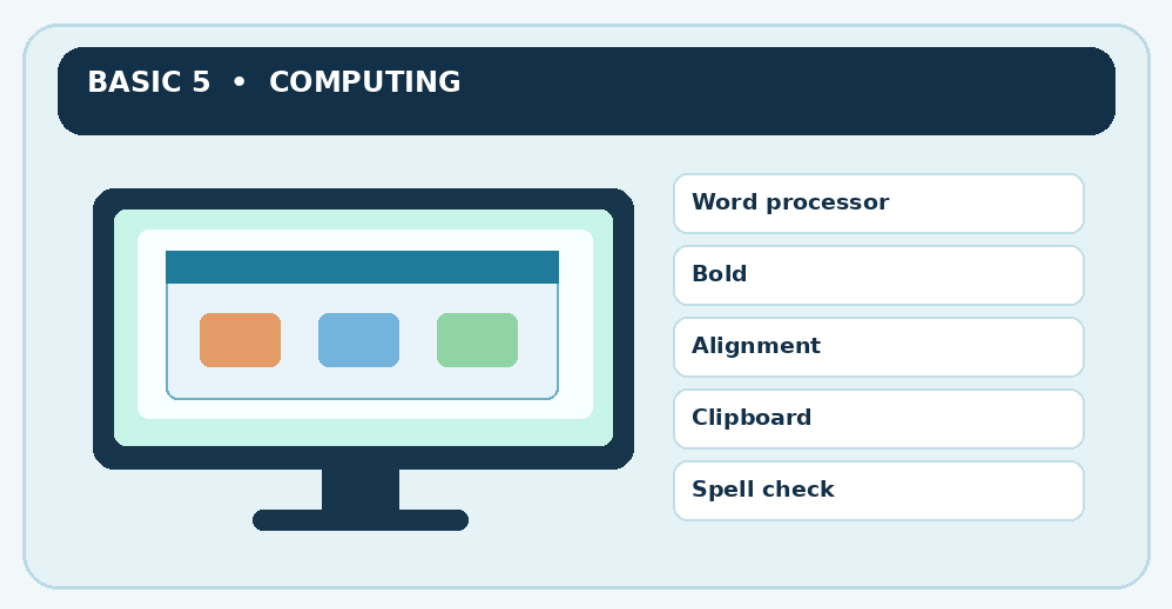
A. Main slide B. Rehearsal C. Image D. Closing slide

10. Present Five Slides — At home, Ama remembers a class rule or concept that sums up the message. What was it called?

A. Opening slide B. Rehearsal C. Main slide D. Closing slide

Lesson 21: Word Processing Ribbons

We will learn to: Identify, explain and apply the important ideas behind word processing ribbons.



Five important ideas

Word processor: creates and edits text documents.

Bold: makes important words stand out.

Alignment: positions text on the page.

Clipboard: temporarily holds copied material.

Spell check: helps find possible spelling mistakes.

Our Ghanaian story

While typing a school announcement, Yaw had a question about word processing ribbons. The group discovered that word processor creates and edits text documents, while bold makes important words stand out. They drew an example and explained it to their classmates.

Try it!

Make a one-page paper example about your school or a Ghanaian festival. Include word processor, bold and alignment; present it clearly to a partner.

Create and share

Make a colourful, labelled poster that teaches word processing ribbons. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.3.1.1.1

Lesson 21: Play & Challenge

PLAY — Choose one best answer for each question.

1. Word Processing Ribbons — Choose the correct name for something that creates and edits text documents.

A. Word processor B. Spell check C. Alignment D. Clipboard

2. Word Processing Ribbons — Your friend says, "This makes important words stand out." What is the friend describing?

A. Alignment B. Bold C. Spell check D. Clipboard

3. Word Processing Ribbons — Which idea from word processing ribbons matches the clue "positions text on the page"?

A. Bold B. Alignment C. Word processor D. Spell check

4. Word Processing Ribbons — The teacher asks for an example that temporarily holds copied material. Which answer fits?

A. Spell check B. Word processor C. Alignment D. Clipboard

5. Word Processing Ribbons — Can you identify the term described here: helps find possible spelling mistakes?

A. Spell check B. Word processor C. Bold D. Alignment

CHALLENGE — Think and apply.

1. Word Processing Ribbons — Your class is solving a practical problem. One useful thing temporarily holds copied material. What is its name?

A. Clipboard B. Alignment C. Word processor D. Spell check

2. Word Processing Ribbons — During a school ICT lesson, Yaw wants to show something that creates and edits text documents. Which idea should be labelled?

A. Clipboard B. Spell check C. Word processor D. Alignment

3. Word Processing Ribbons — A group makes a display on word processing ribbons. Their first clue is "makes important words stand out". Which label is correct?

A. Clipboard B. Bold C. Alignment D. Spell check

4. Word Processing Ribbons — At home, Yaw remembers a class rule or concept that temporarily holds copied material. What was it called?

A. Clipboard B. Spell check C. Alignment D. Word processor

5. Word Processing Ribbons — Which answer would fit on a poster beside the words "creates and edits text documents"?

A. Spell check B. Clipboard C. Word processor D. Alignment

6. Word Processing Ribbons — At the community learning centre, someone describes a tool or idea that helps find possible spelling mistakes. Which one is it?

A. Alignment B. Spell check C. Bold D. Word processor

7. Word Processing Ribbons — Which concept matches the explanation "helps find possible spelling mistakes" when you review word processing ribbons?

A. Bold B. Spell check C. Word processor D. Alignment

8. Word Processing Ribbons — For a short demonstration, the teacher needs an example that positions text on the page. Which choice would you select?

A. Alignment B. Word processor C. Spell check D. Bold

9. Word Processing Ribbons — Imagine helping a younger pupil understand something that positions text on the page. Which term would you teach?

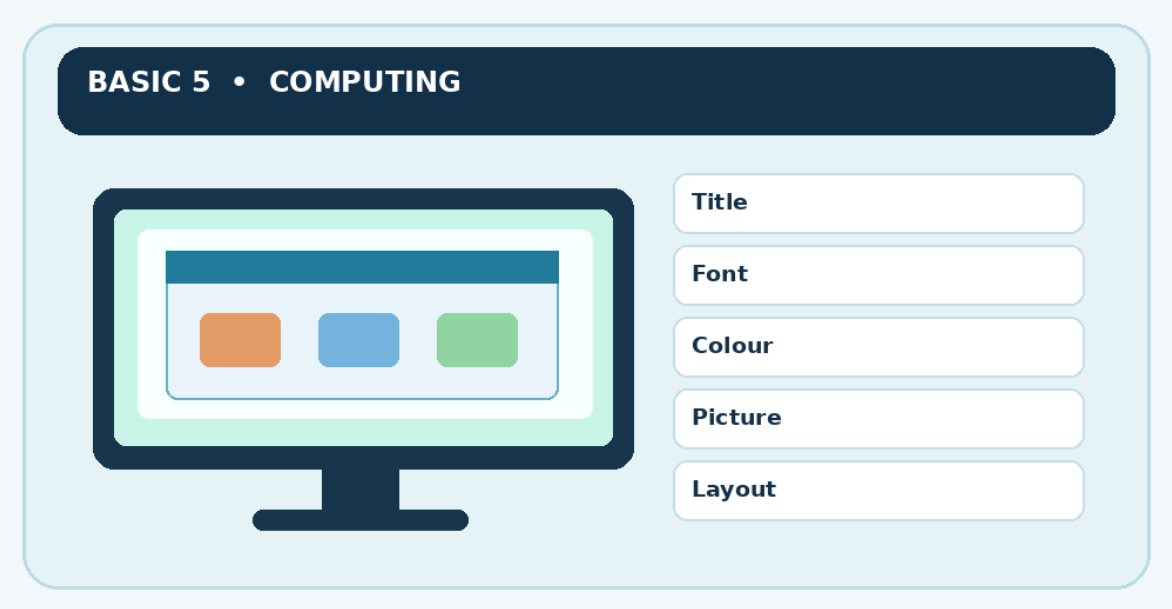
A. Word processor B. Bold C. Alignment D. Spell check

10. Word Processing Ribbons — A classmate confuses several ideas about word processing ribbons. Which one actually makes important words stand out?

A. Alignment B. Bold C. Spell check D. Clipboard

Lesson 22: Design a Well-Formatted Document

We will learn to: Identify, explain and apply the important ideas behind design a well-formatted document.



Five important ideas

Title: states the topic of a page or slide.

Font: changes how writing appears.

Colour: helps emphasise important information.

Picture: can explain an idea visually.

Layout: arranges content clearly.

Our Ghanaian story

While making an Independence Day poster, Abena had a question about design a well-formatted document. The group discovered that title states the topic of a page or slide, while font changes how writing appears. They drew an example and explained it to their classmates.

Try it!

Make a one-page paper example about your school or a Ghanaian festival. Include title, font and colour; present it clearly to a partner.

Create and share

Make a colourful, labelled poster that teaches design a well-formatted document. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.3.1.1.2

Lesson 22: Play & Challenge

PLAY — Choose one best answer for each question.

1. Design a Well-Formatted Document — Choose the correct name for something that states the topic of a page or slide.

A. Picture B. Font C. Layout D. Title

2. Design a Well-Formatted Document — Your friend says, "This changes how writing appears." What is the friend describing?

A. Font B. Picture C. Layout D. Colour

3. Design a Well-Formatted Document — Which idea from design a well-formatted document matches the clue "helps emphasise important information"?

A. Colour B. Font C. Layout D. Picture

4. Design a Well-Formatted Document — The teacher asks for an example that can explain an idea visually. Which answer fits?

A. Picture B. Font C. Layout D. Title

5. Design a Well-Formatted Document — Can you identify the term described here: arranges content clearly?

A. Colour B. Picture C. Font D. Layout

CHALLENGE — Think and apply.

1. Design a Well-Formatted Document — During a school ICT lesson, Kofi wants to show something that states the topic of a page or slide. Which idea should be labelled?

A. Picture B. Layout C. Title D. Font

2. Design a Well-Formatted Document — Which answer would fit on a poster beside the words "states the topic of a page or slide"?

A. Layout B. Picture C. Font D. Title

3. Design a Well-Formatted Document — For a short demonstration, the teacher needs an example that helps emphasise important information. Which choice would you select?

A. Layout B. Picture C. Colour D. Font

4. Design a Well-Formatted Document — Which concept matches the explanation "arranges content clearly" when you review design a well-formatted document?

A. Layout B. Picture C. Colour D. Font

5. Design a Well-Formatted Document — A group makes a display on design a well-formatted document. Their first clue is "changes how writing appears". Which label is correct?

A. Font B. Picture C. Layout D. Colour

6. Design a Well-Formatted Document — At the community learning centre, someone describes a tool or idea that arranges content clearly. Which one is it?

A. Picture B. Font C. Layout D. Colour

7. Design a Well-Formatted Document — A classmate confuses several ideas about design a well-formatted document. Which one actually changes how writing appears?

A. Colour B. Font C. Picture D. Layout

8. Design a Well-Formatted Document — Your class is solving a practical problem. One useful thing can explain an idea visually. What is its name?

A. Picture B. Font C. Title D. Layout

9. Design a Well-Formatted Document — At home, Kofi remembers a class rule or concept that can explain an idea visually. What was it called?

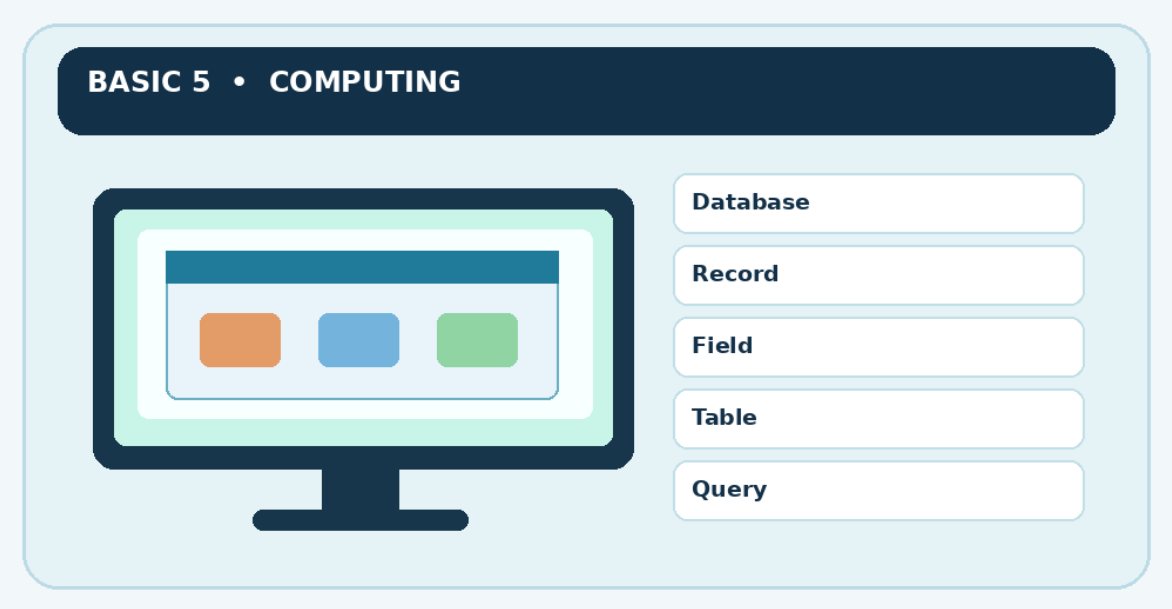
A. Layout B. Title C. Picture D. Font

10. Design a Well-Formatted Document — Imagine helping a younger pupil understand something that helps emphasise important information. Which term would you teach?

A. Font B. Colour C. Picture D. Layout

Lesson 23: Introducing Databases

We will learn to: Identify, explain and apply the important ideas behind introducing databases.



Five important ideas

Database: is an organised collection of related information.

Record: is one complete entry.

Field: is a named piece of data in an entry.

Table: arranges records in rows and fields in columns.

Query: asks a database to find matching information.

Our Ghanaian story

When keeping a register of library books, Kofi had a question about introducing databases. The group discovered that database is an organised collection of related information, while record is one complete entry. They drew an example and explained it to their classmates.

Try it!

Collect a few harmless facts about classroom materials, NOT personal details. Make a paper chart with the terms Database, Record and Field; explain what each means.

Create and share

Make a colourful, labelled poster that teaches introducing databases. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.5.1.3.1

Lesson 23: Play & Challenge

PLAY — Choose one best answer for each question.

1. Introducing Databases — Choose the correct name for something that is an organised collection of related information.

A. Query B. Table C. Database D. Record

2. Introducing Databases — Your friend says, "This is one complete entry." What is the friend describing?

A. Query B. Database C. Table D. Record

3. Introducing Databases — Which idea from introducing databases matches the clue "is a named piece of data in an entry"?

A. Query B. Field C. Table D. Record

4. Introducing Databases — The teacher asks for an example that arranges records in rows and fields in columns. Which answer fits?

A. Query B. Database C. Field D. Table

5. Introducing Databases — Can you identify the term described here: asks a database to find matching information?

A. Field B. Query C. Database D. Record

CHALLENGE — Think and apply.

1. Introducing Databases — At home, Akosua remembers a class rule or concept that arranges records in rows and fields in columns. What was it called?

A. Table B. Query C. Field D. Database

2. Introducing Databases — For a short demonstration, the teacher needs an example that is a named piece of data in an entry. Which choice would you select?

A. Table B. Field C. Query D. Record

3. Introducing Databases — Your class is solving a practical problem. One useful thing arranges records in rows and fields in columns. What is its name?

A. Query B. Field C. Table D. Database

4. Introducing Databases — During a school ICT lesson, Akosua wants to show something that is an organised collection of related information. Which idea should be labelled?

A. Table B. Query C. Record D. Database

5. Introducing Databases — Which concept matches the explanation "asks a database to find matching information" when you review introducing databases?

A. Database B. Record C. Query D. Field

6. Introducing Databases — Which answer would fit on a poster beside the words "is an organised collection of related information"?

A. Record B. Query C. Table D. Database

7. Introducing Databases — A group makes a display on introducing databases. Their first clue is "is one complete entry". Which label is correct?

A. Database B. Record C. Query D. Table

8. Introducing Databases — Imagine helping a younger pupil understand something that is a named piece of data in an entry. Which term would you teach?

A. Query B. Table C. Record D. Field

9. Introducing Databases — A classmate confuses several ideas about introducing databases. Which one actually is one complete entry?

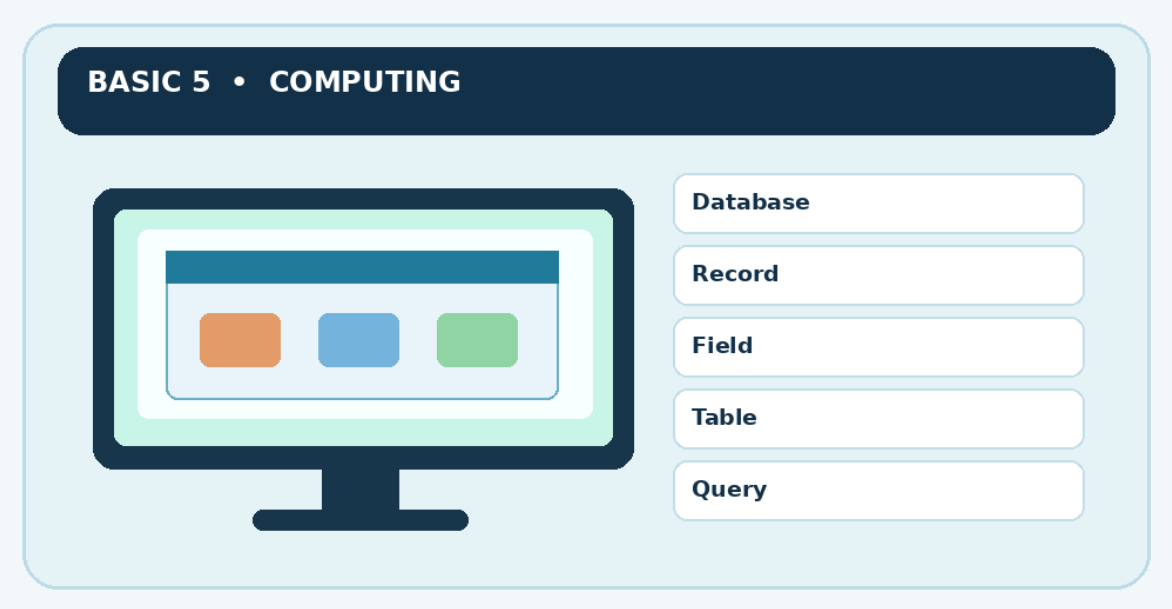
A. Database B. Record C. Query D. Table

10. Introducing Databases — At the community learning centre, someone describes a tool or idea that asks a database to find matching information. Which one is it?

A. Database B. Query C. Field D. Record

Lesson 24: Why We Use Databases

We will learn to: Identify, explain and apply the important ideas behind why we use databases.



Five important ideas

Database: is an organised collection of related information.

Record: is one complete entry.

Field: is a named piece of data in an entry.

Table: arranges records in rows and fields in columns.

Query: asks a database to find matching information.

Our Ghanaian story

When keeping a register of library books, Ama had a question about why we use databases. The group discovered that database is an organised collection of related information, while record is one complete entry. They drew an example and explained it to their classmates.

Try it!

Collect a few harmless facts about classroom materials, NOT personal details. Make a paper chart with the terms Database, Record and Field; explain what each means.

Create and share

Make a colourful, labelled poster that teaches why we use databases. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.5.1.3.2

Lesson 24: Play & Challenge

PLAY — Choose one best answer for each question.

1. Why We Use Databases — Choose the correct name for something that is an organised collection of related information.

A. Query B. Database C. Record D. Field

2. Why We Use Databases — Your friend says, "This is one complete entry." What is the friend describing?

A. Table B. Query C. Record D. Field

3. Why We Use Databases — Which idea from why we use databases matches the clue "is a named piece of data in an entry"?

A. Table B. Field C. Query D. Database

4. Why We Use Databases — The teacher asks for an example that arranges records in rows and fields in columns. Which answer fits?

A. Query B. Field C. Database D. Table

5. Why We Use Databases — Can you identify the term described here: asks a database to find matching information?

A. Record B. Field C. Query D. Table

CHALLENGE — Think and apply.

1. Why We Use Databases — At the community learning centre, someone describes a tool or idea that asks a database to find matching information. Which one is it?

A. Record B. Query C. Table D. Field

2. Why We Use Databases — A group makes a display on why we use databases. Their first clue is "is one complete entry". Which label is correct?

A. Record B. Field C. Query D. Table

3. Why We Use Databases — During a school ICT lesson, Esi wants to show something that is an organised collection of related information. Which idea should be labelled?

A. Field B. Record C. Query D. Database

4. Why We Use Databases — Imagine helping a younger pupil understand something that is a named piece of data in an entry. Which term would you teach?

A. Table B. Query C. Database D. Field

5. Why We Use Databases — At home, Esi remembers a class rule or concept that arranges records in rows and fields in columns. What was it called?

A. Table B. Database C. Query D. Field

6. Why We Use Databases — Which answer would fit on a poster beside the words "is an organised collection of related information"?

A. Database B. Query C. Record D. Field

7. Why We Use Databases — For a short demonstration, the teacher needs an example that is a named piece of data in an entry. Which choice would you select?

A. Query B. Table C. Database D. Field

8. Why We Use Databases — A classmate confuses several ideas about why we use databases. Which one actually is one complete entry?

A. Query B. Field C. Table D. Record

9. Why We Use Databases — Which concept matches the explanation "asks a database to find matching information" when you review why we use databases?

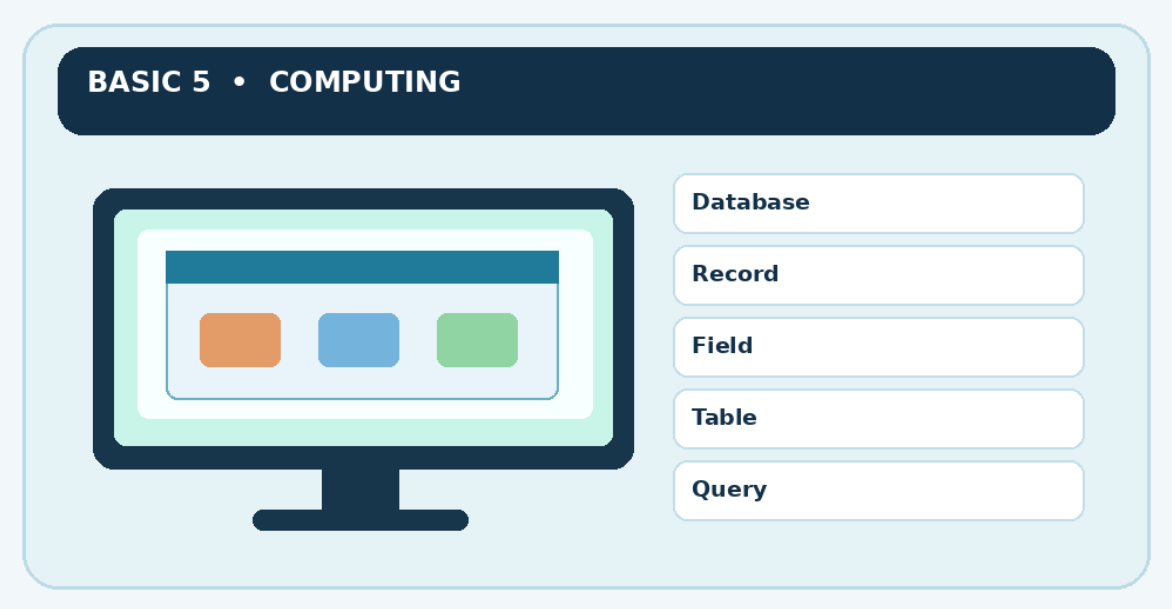
A. Query B. Record C. Table D. Field

10. Why We Use Databases — Your class is solving a practical problem. One useful thing arranges records in rows and fields in columns. What is its name?

A. Query B. Field C. Database D. Table

Lesson 25: Database Tables and Records

We will learn to: Identify, explain and apply the important ideas behind database tables and records.



Five important ideas

Database: is an organised collection of related information.

Record: is one complete entry.

Field: is a named piece of data in an entry.

Table: arranges records in rows and fields in columns.

Query: asks a database to find matching information.

Our Ghanaian story

When keeping a register of library books, Kojo had a question about database tables and records. The group discovered that database is an organised collection of related information, while record is one complete entry. They drew an example and explained it to their classmates.

Try it!

Collect a few harmless facts about classroom materials, NOT personal details. Make a paper chart with the terms Database, Record and Field; explain what each means.

Create and share

Make a colourful, labelled poster that teaches database tables and records. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.5.1.3.5

Lesson 25: Play & Challenge

PLAY — Choose one best answer for each question.

1. Database Tables and Records — Choose the correct name for something that is an organised collection of related information.

A. Query B. Database C. Table D. Record

2. Database Tables and Records — Your friend says, "This is one complete entry." What is the friend describing?

A. Table B. Record C. Field D. Query

3. Database Tables and Records — Which idea from database tables and records matches the clue "is a named piece of data in an entry"?

A. Query B. Table C. Field D. Database

4. Database Tables and Records — The teacher asks for an example that arranges records in rows and fields in columns. Which answer fits?

A. Record B. Field C. Database D. Table

5. Database Tables and Records — Can you identify the term described here: asks a database to find matching information?

A. Record B. Table C. Field D. Query

CHALLENGE — Think and apply.

1. Database Tables and Records — Your class is solving a practical problem. One useful thing arranges records in rows and fields in columns. What is its name?

A. Field B. Database C. Table D. Record

2. Database Tables and Records — Which concept matches the explanation "asks a database to find matching information" when you review database tables and records?

A. Query B. Field C. Record D. Table

3. Database Tables and Records — During a school ICT lesson, Ama wants to show something that is an organised collection of related information. Which idea should be labelled?

A. Record B. Database C. Query D. Table

4. Database Tables and Records — A group makes a display on database tables and records. Their first clue is "is one complete entry". Which label is correct?

A. Record B. Table C. Field D. Query

5. Database Tables and Records — At home, Ama remembers a class rule or concept that arranges records in rows and fields in columns. What was it called?

A. Field B. Database C. Record D. Table

6. Database Tables and Records — Imagine helping a younger pupil understand something that is a named piece of data in an entry. Which term would you teach?

A. Table B. Query C. Database D. Field

7. Database Tables and Records — A classmate confuses several ideas about database tables and records. Which one actually is one complete entry?

A. Query B. Table C. Record D. Field

8. Database Tables and Records — At the community learning centre, someone describes a tool or idea that asks a database to find matching information. Which one is it?

A. Table B. Field C. Query D. Record

9. Database Tables and Records — For a short demonstration, the teacher needs an example that is a named piece of data in an entry. Which choice would you select?

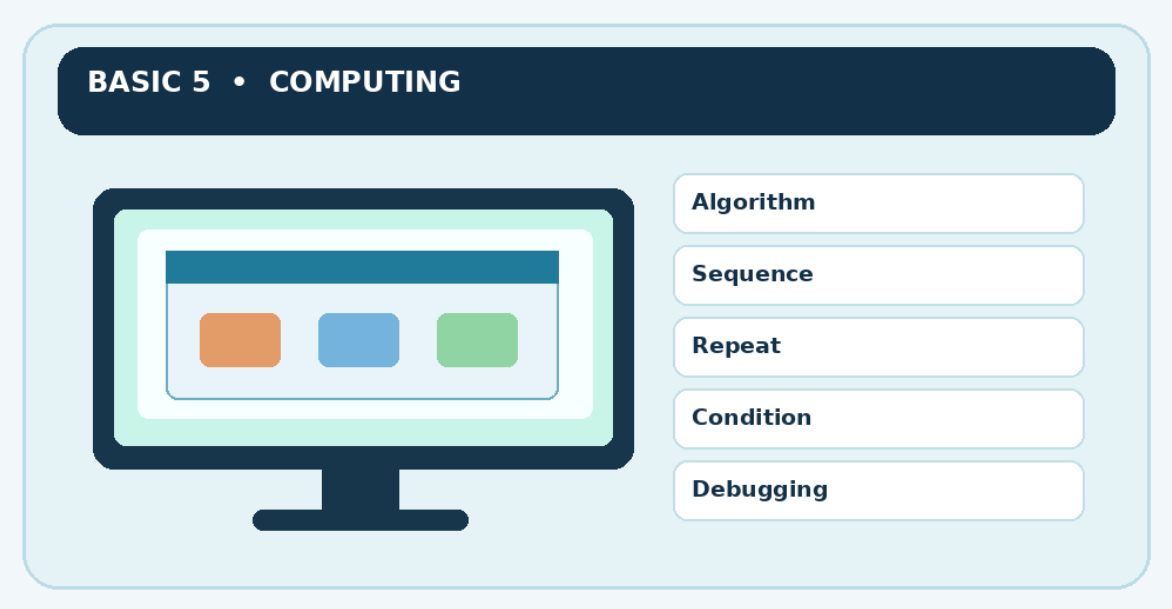
A. Field B. Query C. Database D. Table

10. Database Tables and Records — Which answer would fit on a poster beside the words "is an organised collection of related information"?

A. Table B. Database C. Query D. Record

Lesson 26: Programming Languages

We will learn to: Identify, explain and apply the important ideas behind programming languages.



Five important ideas

Algorithm: is an ordered set of steps to solve a task.

Sequence: puts instructions in the right order.

Repeat: does an instruction again.

Condition: chooses an action based on a test.

Debugging: looks for and fixes errors in instructions.

Our Ghanaian story

When giving a robot simple directions, Esi had a question about programming languages. The group discovered that algorithm is an ordered set of steps to solve a task, while sequence puts instructions in the right order. They drew an example and explained it to their classmates.

Try it!

Design a simple paper model explaining algorithm, sequence and repeat. Trade models with a classmate and explain how your design works.

Create and share

Make a colourful, labelled poster that teaches programming languages. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.5.1.3.6

Lesson 26: Play & Challenge

PLAY — Choose one best answer for each question.

1. Programming Languages — Choose the correct name for something that is an ordered set of steps to solve a task.

A. Condition B. Sequence C. Algorithm D. Repeat

2. Programming Languages — Your friend says, "This puts instructions in the right order." What is the friend describing?

A. Condition B. Sequence C. Algorithm D. Debugging

3. Programming Languages — Which idea from programming languages matches the clue "does an instruction again"?

A. Condition B. Debugging C. Repeat D. Sequence

4. Programming Languages — The teacher asks for an example that chooses an action based on a test. Which answer fits?

A. Algorithm B. Repeat C. Debugging D. Condition

5. Programming Languages — Can you identify the term described here: looks for and fixes errors in instructions?

A. Condition B. Repeat C. Algorithm D. Debugging

CHALLENGE — Think and apply.

1. Programming Languages — Which concept matches the explanation "looks for and fixes errors in instructions" when you review programming languages?

A. Repeat B. Debugging C. Algorithm D. Condition

2. Programming Languages — Which answer would fit on a poster beside the words "is an ordered set of steps to solve a task"?

A. Sequence B. Repeat C. Condition D. Algorithm

3. Programming Languages — A classmate confuses several ideas about programming languages. Which one actually puts instructions in the right order?

A. Algorithm B. Sequence C. Condition D. Debugging

4. Programming Languages — At the community learning centre, someone describes a tool or idea that looks for and fixes errors in instructions. Which one is it?

A. Debugging B. Repeat C. Algorithm D. Condition

5. Programming Languages — During a school ICT lesson, Yaw wants to show something that is an ordered set of steps to solve a task. Which idea should be labelled?

A. Algorithm B. Sequence C. Condition D. Repeat

6. Programming Languages — For a short demonstration, the teacher needs an example that does an instruction again. Which choice would you select?

A. Condition B. Sequence C. Debugging D. Repeat

7. Programming Languages — Your class is solving a practical problem. One useful thing chooses an action based on a test. What is its name?

A. Algorithm B. Debugging C. Repeat D. Condition

8. Programming Languages — A group makes a display on programming languages. Their first clue is "puts instructions in the right order". Which label is correct?

A. Sequence B. Condition C. Algorithm D. Debugging

9. Programming Languages — Imagine helping a younger pupil understand something that does an instruction again. Which term would you teach?

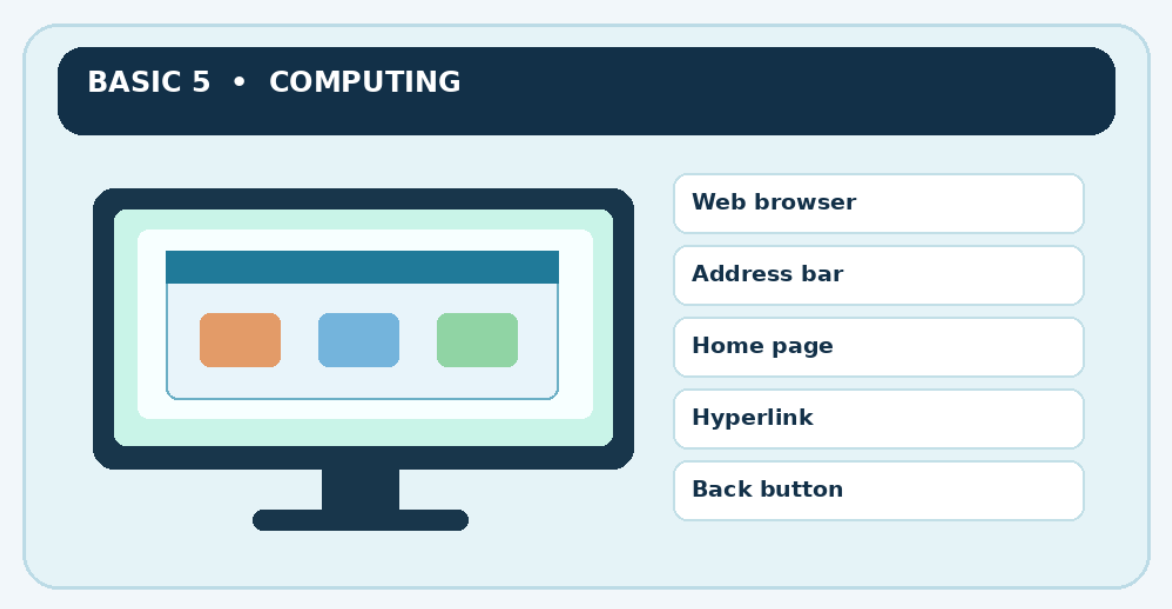
A. Repeat B. Debugging C. Condition D. Sequence

10. Programming Languages — At home, Yaw remembers a class rule or concept that chooses an action based on a test. What was it called?

A. Algorithm B. Debugging C. Condition D. Repeat

Lesson 27: Understanding Web Browsers

We will learn to: Identify, explain and apply the important ideas behind understanding web browsers.



Five important ideas

Web browser: opens websites and web pages.

Address bar: shows or accepts a website address.

Home page: is the main starting page of a website.

Hyperlink: opens another page or resource when selected.

Back button: returns to a previously visited page.

Our Ghanaian story

While exploring a teacher-approved website, Yaw had a question about understanding web browsers. The group discovered that web browser opens websites and web pages, while address bar shows or accepts a website address. They drew an example and explained it to their classmates.

Try it!

Use an imaginary example or a teacher-approved offline screenshot. Discuss when to use web browser and address bar. Never enter a real password or share private details in this activity.

Create and share

Make a colourful, labelled poster that teaches understanding web browsers. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.6.2.1.1

Lesson 27: Play & Challenge

PLAY — Choose one best answer for each question.

1. Understanding Web Browsers — Choose the correct name for something that opens websites and web pages.

A. Address bar B. Web browser C. Back button D. Hyperlink

2. Understanding Web Browsers — Your friend says, "This shows or accepts a website address." What is the friend describing?

A. Web browser B. Back button C. Hyperlink D. Address bar

3. Understanding Web Browsers — Which idea from understanding web browsers matches the clue "is the main starting page of a website"?

A. Back button B. Home page C. Web browser D. Hyperlink

4. Understanding Web Browsers — The teacher asks for an example that opens another page or resource when selected. Which answer fits?

A. Hyperlink B. Web browser C. Back button D. Home page

5. Understanding Web Browsers — Can you identify the term described here: returns to a previously visited page?

A. Address bar B. Home page C. Web browser D. Back button

CHALLENGE — Think and apply.

1. Understanding Web Browsers — During a school ICT lesson, Kofi wants to show something that opens websites and web pages. Which idea should be labelled?

A. Web browser B. Hyperlink C. Back button D. Address bar

2. Understanding Web Browsers — A group makes a display on understanding web browsers. Their first clue is "shows or accepts a website address". Which label is correct?

A. Back button B. Hyperlink C. Web browser D. Address bar

3. Understanding Web Browsers — Which answer would fit on a poster beside the words "opens websites and web pages"?

A. Web browser B. Back button C. Address bar D. Hyperlink

4. Understanding Web Browsers — Imagine helping a younger pupil understand something that is the main starting page of a website. Which term would you teach?

A. Hyperlink B. Home page C. Back button D. Web browser

5. Understanding Web Browsers — At home, Kofi remembers a class rule or concept that opens another page or resource when selected. What was it called?

A. Web browser B. Hyperlink C. Back button D. Home page

6. Understanding Web Browsers — Which concept matches the explanation "returns to a previously visited page" when you review understanding web browsers?

A. Home page B. Web browser C. Address bar D. Back button

7. Understanding Web Browsers — For a short demonstration, the teacher needs an example that is the main starting page of a website. Which choice would you select?

A. Hyperlink B. Home page C. Back button D. Web browser

8. Understanding Web Browsers — Your class is solving a practical problem. One useful thing opens another page or resource when selected. What is its name?

A. Back button B. Hyperlink C. Web browser D. Home page

9. Understanding Web Browsers — At the community learning centre, someone describes a tool or idea that returns to a previously visited page. Which one is it?

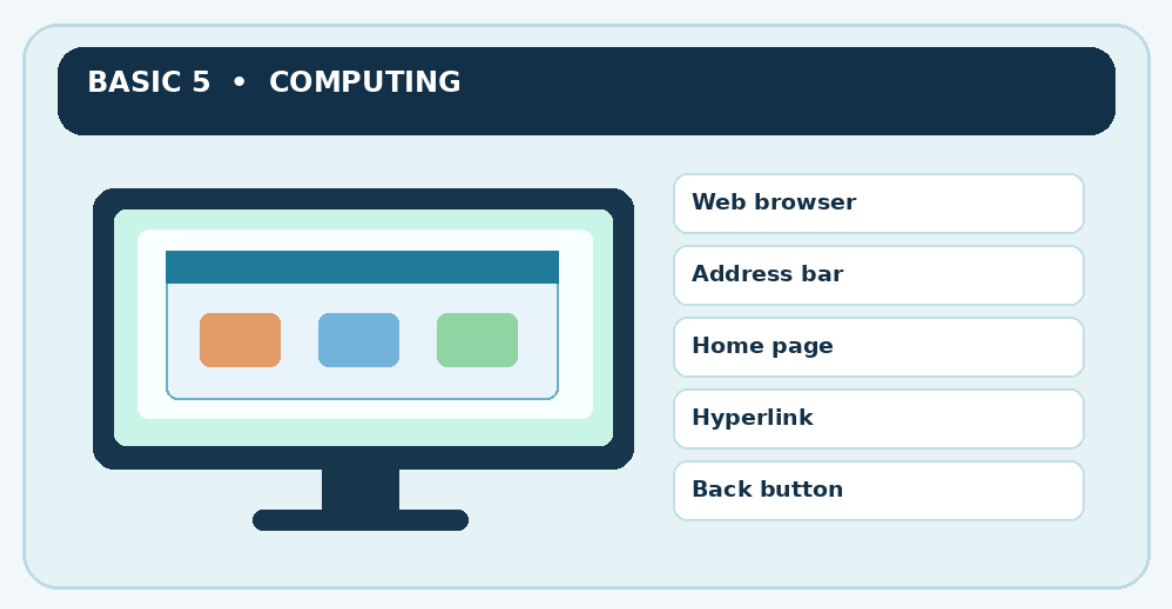
A. Home page B. Web browser C. Back button D. Address bar

10. Understanding Web Browsers — A classmate confuses several ideas about understanding web browsers. Which one actually shows or accepts a website address?

A. Address bar B. Web browser C. Hyperlink D. Back button

Lesson 28: What Is a Web Page?

We will learn to: Identify, explain and apply the important ideas behind what is a web page?.



Five important ideas

Web browser: opens websites and web pages.

Address bar: shows or accepts a website address.

Home page: is the main starting page of a website.

Hyperlink: opens another page or resource when selected.

Back button: returns to a previously visited page.

Our Ghanaian story

While exploring a teacher-approved website, Abena had a question about what is a web page?. The group discovered that web browser opens websites and web pages, while address bar shows or accepts a website address. They drew an example and explained it to their classmates.

Try it!

Use an imaginary example or a teacher-approved offline screenshot. Discuss when to use web browser and address bar. Never enter a real password or share private details in this activity.

Create and share

Make a colourful, labelled poster that teaches what is a web page?. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.6.2.1.7

Lesson 28: Play & Challenge

PLAY — Choose one best answer for each question.

1. What Is a Web Page? — Choose the correct name for something that opens websites and web pages.

A. Hyperlink B. Home page C. Web browser D. Address bar

2. What Is a Web Page? — Your friend says, "This shows or accepts a website address." What is the friend describing?

A. Address bar B. Hyperlink C. Home page D. Web browser

3. What Is a Web Page? — Which idea from what is a web page? matches the clue "is the main starting page of a website"?

A. Home page B. Address bar C. Hyperlink D. Back button

4. What Is a Web Page? — The teacher asks for an example that opens another page or resource when selected. Which answer fits?

A. Back button B. Address bar C. Home page D. Hyperlink

5. What Is a Web Page? — Can you identify the term described here: returns to a previously visited page?

A. Address bar B. Web browser C. Back button D. Home page

CHALLENGE — Think and apply.

1. What Is a Web Page? — Your class is solving a practical problem. One useful thing opens another page or resource when selected. What is its name?

A. Hyperlink B. Back button C. Address bar D. Home page

2. What Is a Web Page? — A group makes a display on what is a web page?. Their first clue is "shows or accepts a website address". Which label is correct?

A. Address bar B. Hyperlink C. Home page D. Web browser

3. What Is a Web Page? — A classmate confuses several ideas about what is a web page?. Which one actually shows or accepts a website address?

A. Web browser B. Address bar C. Hyperlink D. Home page

4. What Is a Web Page? — At home, Akosua remembers a class rule or concept that opens another page or resource when selected. What was it called?

A. Home page B. Address bar C. Back button D. Hyperlink

5. What Is a Web Page? — At the community learning centre, someone describes a tool or idea that returns to a previously visited page. Which one is it?

A. Address bar B. Home page C. Web browser D. Back button

6. What Is a Web Page? — For a short demonstration, the teacher needs an example that is the main starting page of a website. Which choice would you select?

A. Hyperlink B. Home page C. Back button D. Address bar

7. What Is a Web Page? — During a school ICT lesson, Akosua wants to show something that opens websites and web pages. Which idea should be labelled?

A. Hyperlink B. Address bar C. Web browser D. Home page

8. What Is a Web Page? — Which answer would fit on a poster beside the words "opens websites and web pages"?

A. Address bar B. Hyperlink C. Web browser D. Home page

9. What Is a Web Page? — Imagine helping a younger pupil understand something that is the main starting page of a website. Which term would you teach?

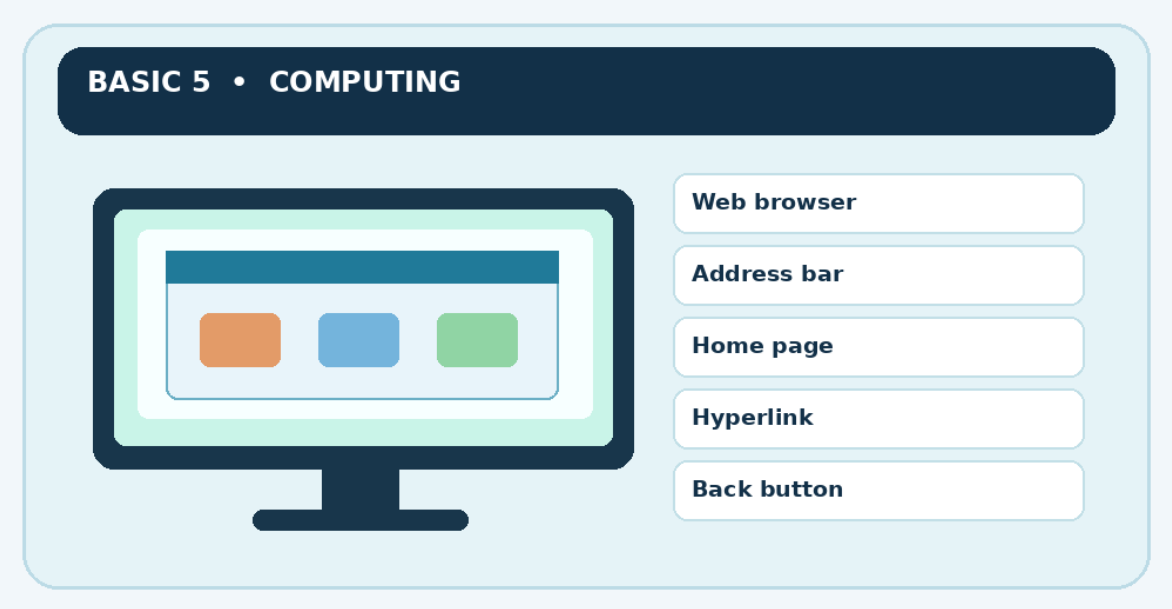
A. Address bar B. Hyperlink C. Back button D. Home page

10. What Is a Web Page? — Which concept matches the explanation "returns to a previously visited page" when you review what is a web page??

A. Web browser B. Address bar C. Home page D. Back button

Lesson 29: Following Hyperlinks

We will learn to: Identify, explain and apply the important ideas behind following hyperlinks.



Five important ideas

Web browser: opens websites and web pages.

Address bar: shows or accepts a website address.

Home page: is the main starting page of a website.

Hyperlink: opens another page or resource when selected.

Back button: returns to a previously visited page.

Our Ghanaian story

While exploring a teacher-approved website, Kofi had a question about following hyperlinks. The group discovered that web browser opens websites and web pages, while address bar shows or accepts a website address. They drew an example and explained it to their classmates.

Try it!

Use an imaginary example or a teacher-approved offline screenshot. Discuss when to use web browser and address bar. Never enter a real password or share private details in this activity.

Create and share

Make a colourful, labelled poster that teaches following hyperlinks. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.6.2.1.9

Lesson 29: Play & Challenge

PLAY — Choose one best answer for each question.

1. Following Hyperlinks — Choose the correct name for something that opens websites and web pages.

A. Web browser B. Address bar C. Back button D. Home page

2. Following Hyperlinks — Your friend says, "This shows or accepts a website address." What is the friend describing?

A. Web browser B. Hyperlink C. Address bar D. Back button

3. Following Hyperlinks — Which idea from following hyperlinks matches the clue "is the main starting page of a website"?

A. Web browser B. Home page C. Hyperlink D. Back button

4. Following Hyperlinks — The teacher asks for an example that opens another page or resource when selected. Which answer fits?

A. Web browser B. Back button C. Home page D. Hyperlink

5. Following Hyperlinks — Can you identify the term described here: returns to a previously visited page?

A. Address bar B. Back button C. Hyperlink D. Home page

CHALLENGE — Think and apply.

1. Following Hyperlinks — A group makes a display on following hyperlinks. Their first clue is "shows or accepts a website address". Which label is correct?

A. Web browser B. Back button C. Hyperlink D. Address bar

2. Following Hyperlinks — Which answer would fit on a poster beside the words "opens websites and web pages"?

A. Back button B. Address bar C. Home page D. Web browser

3. Following Hyperlinks — For a short demonstration, the teacher needs an example that is the main starting page of a website. Which choice would you select?

A. Web browser B. Hyperlink C. Back button D. Home page

4. Following Hyperlinks — Which concept matches the explanation "returns to a previously visited page" when you review following hyperlinks?

A. Back button B. Home page C. Hyperlink D. Address bar

5. Following Hyperlinks — At home, Esi remembers a class rule or concept that opens another page or resource when selected. What was it called?

A. Home page B. Hyperlink C. Web browser D. Back button

6. Following Hyperlinks — During a school ICT lesson, Esi wants to show something that opens websites and web pages. Which idea should be labelled?

A. Web browser B. Address bar C. Back button D. Home page

7. Following Hyperlinks — At the community learning centre, someone describes a tool or idea that returns to a previously visited page. Which one is it?

A. Address bar B. Home page C. Hyperlink D. Back button

8. Following Hyperlinks — Imagine helping a younger pupil understand something that is the main starting page of a website. Which term would you teach?

A. Hyperlink B. Web browser C. Home page D. Back button

9. Following Hyperlinks — A classmate confuses several ideas about following hyperlinks. Which one actually shows or accepts a website address?

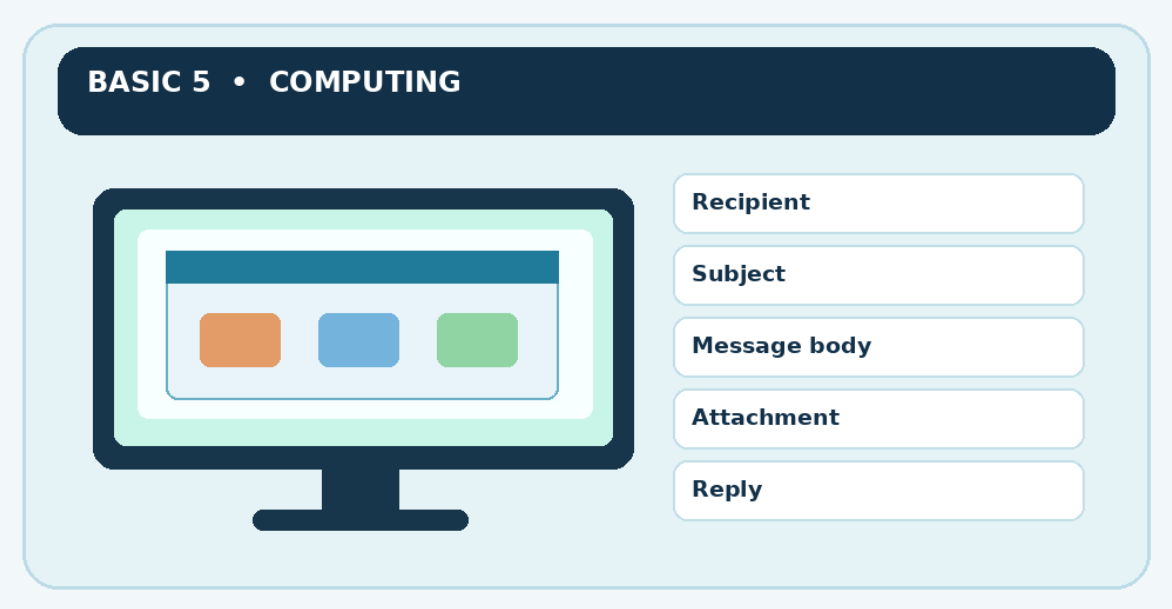
A. Back button B. Hyperlink C. Web browser D. Address bar

10. Following Hyperlinks — Your class is solving a practical problem. One useful thing opens another page or resource when selected. What is its name?

A. Home page B. Back button C. Hyperlink D. Web browser

Lesson 30: Email Services

We will learn to: Identify, explain and apply the important ideas behind email services.



Five important ideas

Recipient: is the person meant to receive an email.

Subject: says briefly what the message is about.

Message body: contains the main written information.

Attachment: is a file sent with a message.

Reply: sends a response to a received message.

Our Ghanaian story

When writing a supervised class message, Ama had a question about email services. The group discovered that recipient is the person meant to receive an email, while subject says briefly what the message is about. They drew an example and explained it to their classmates.

Try it!

Use an imaginary example or a teacher-approved offline screenshot. Discuss when to use recipient and subject. Never enter a real password or share private details in this activity.

Create and share

Make a colourful, labelled poster that teaches email services. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.6.7.1.1

Lesson 30: Play & Challenge

PLAY — Choose one best answer for each question.

1. Email Services — Choose the correct name for something that is the person meant to receive an email.

A. Reply B. Recipient C. Attachment D. Subject

2. Email Services — Your friend says, "This says briefly what the message is about." What is the friend describing?

A. Reply B. Subject C. Message body D. Recipient

3. Email Services — Which idea from email services matches the clue "contains the main written information"?

A. Reply B. Subject C. Attachment D. Message body

4. Email Services — The teacher asks for an example that is a file sent with a message. Which answer fits?

A. Attachment B. Message body C. Recipient D. Reply

5. Email Services — Can you identify the term described here: sends a response to a received message?

A. Subject B. Message body C. Reply D. Recipient

CHALLENGE — Think and apply.

1. Email Services — A classmate confuses several ideas about email services. Which one actually says briefly what the message is about?

A. Recipient B. Message body C. Subject D. Reply

2. Email Services — At the community learning centre, someone describes a tool or idea that sends a response to a received message. Which one is it?

A. Reply B. Subject C. Recipient D. Message body

3. Email Services — For a short demonstration, the teacher needs an example that contains the main written information. Which choice would you select?

A. Message body B. Reply C. Subject D. Attachment

4. Email Services — During a school ICT lesson, Ama wants to show something that is the person meant to receive an email. Which idea should be labelled?

A. Subject B. Reply C. Attachment D. Recipient

5. Email Services — Which concept matches the explanation "sends a response to a received message" when you review email services?

A. Recipient B. Message body C. Subject D. Reply

6. Email Services — At home, Ama remembers a class rule or concept that is a file sent with a message. What was it called?

A. Attachment B. Message body C. Reply D. Recipient

7. Email Services — Imagine helping a younger pupil understand something that contains the main written information. Which term would you teach?

A. Subject B. Attachment C. Reply D. Message body

8. Email Services — A group makes a display on email services. Their first clue is "says briefly what the message is about". Which label is correct?

A. Message body B. Recipient C. Subject D. Reply

9. Email Services — Which answer would fit on a poster beside the words "is the person meant to receive an email"?

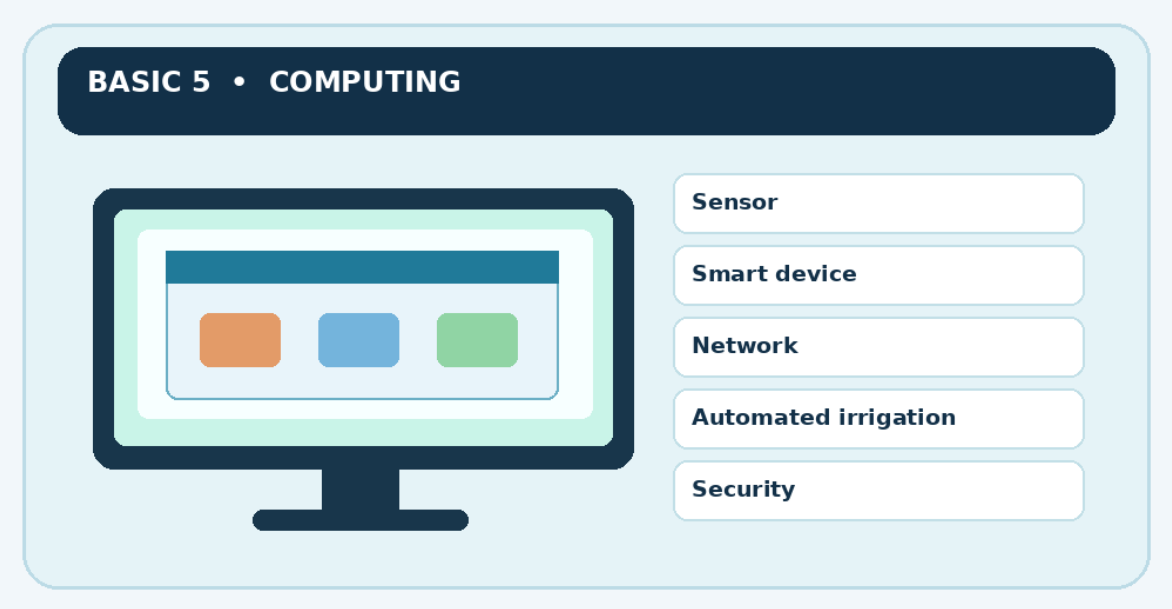
A. Attachment B. Subject C. Recipient D. Reply

10. Email Services — Your class is solving a practical problem. One useful thing is a file sent with a message. What is its name?

A. Reply B. Message body C. Recipient D. Attachment

Lesson 31: Internet of Things

We will learn to: Identify, explain and apply the important ideas behind internet of things.



Five important ideas

Sensor: detects a change such as temperature or movement.

Smart device: can use connected data to act.

Network: allows devices to exchange information.

Automated irrigation: can water crops based on programmed rules.

Security: protects connected devices and their information.

Our Ghanaian story

At a demonstration of a smart farming system, Kojo had a question about internet of things. The group discovered that sensor detects a change such as temperature or movement, while smart device can use connected data to act. They drew an example and explained it to their classmates.

Try it!

Design a simple paper model explaining sensor, smart device and network. Trade models with a classmate and explain how your design works.

Create and share

Make a colourful, labelled poster that teaches internet of things. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.6.8.1.1

Lesson 31: Play & Challenge

PLAY — Choose one best answer for each question.

1. Internet of Things — Choose the correct name for something that detects a change such as temperature or movement.

A. Sensor B. Security C. Network D. Automated irrigation

2. Internet of Things — Your friend says, "This can use connected data to act." What is the friend describing?

A. Automated irrigation B. Security C. Network D. Smart device

3. Internet of Things — Which idea from internet of things matches the clue "allows devices to exchange information"?

A. Sensor B. Smart device C. Automated irrigation D. Network

4. Internet of Things — The teacher asks for an example that can water crops based on programmed rules. Which answer fits?

A. Sensor B. Automated irrigation C. Smart device D. Security

5. Internet of Things — Can you identify the term described here: protects connected devices and their information?

A. Sensor B. Security C. Smart device D. Network

CHALLENGE — Think and apply.

1. Internet of Things — Which concept matches the explanation "protects connected devices and their information" when you review internet of things?

A. Security B. Sensor C. Network D. Smart device

2. Internet of Things — At the community learning centre, someone describes a tool or idea that protects connected devices and their information. Which one is it?

A. Network B. Sensor C. Smart device D. Security

3. Internet of Things — Imagine helping a younger pupil understand something that allows devices to exchange information. Which term would you teach?

A. Network B. Smart device C. Automated irrigation D. Sensor

4. Internet of Things — During a school ICT lesson, Yaw wants to show something that detects a change such as temperature or movement. Which idea should be labelled?

A. Network B. Automated irrigation C. Sensor D. Security

5. Internet of Things — A classmate confuses several ideas about internet of things. Which one actually can use connected data to act?

A. Security B. Smart device C. Automated irrigation D. Network

6. Internet of Things — Your class is solving a practical problem. One useful thing can water crops based on programmed rules. What is its name?

A. Sensor B. Automated irrigation C. Smart device D. Security

7. Internet of Things — A group makes a display on internet of things. Their first clue is "can use connected data to act". Which label is correct?

A. Automated irrigation B. Smart device C. Security D. Network

8. Internet of Things — Which answer would fit on a poster beside the words "detects a change such as temperature or movement"?

A. Security B. Automated irrigation C. Network D. Sensor

9. Internet of Things — For a short demonstration, the teacher needs an example that allows devices to exchange information. Which choice would you select?

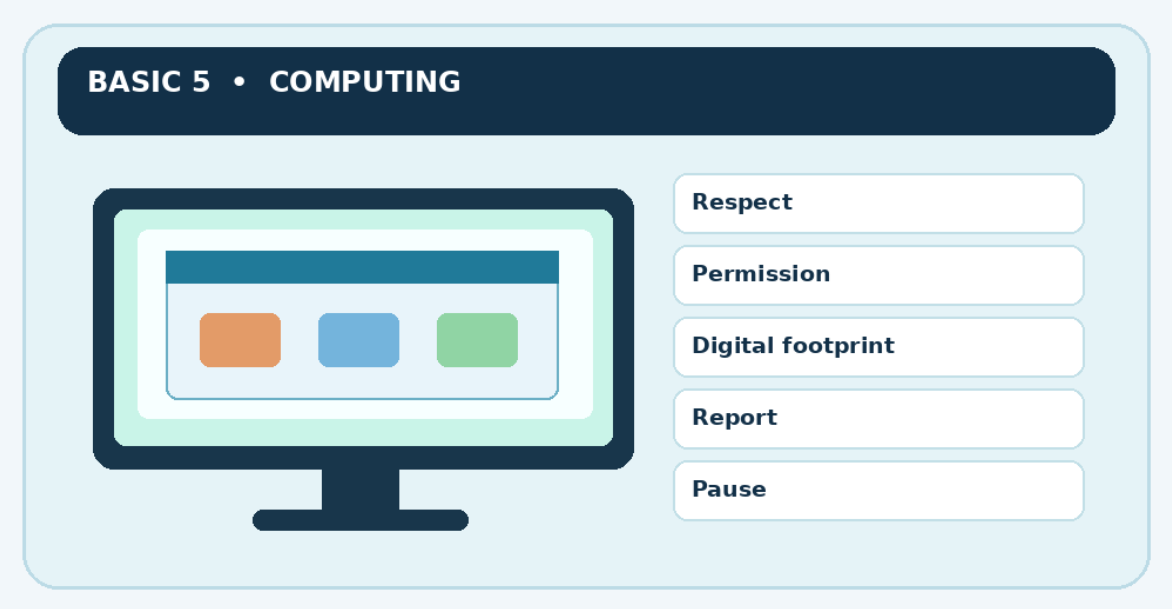
A. Smart device B. Network C. Automated irrigation D. Sensor

10. Internet of Things — At home, Yaw remembers a class rule or concept that can water crops based on programmed rules. What was it called?

A. Security B. Automated irrigation C. Smart device D. Sensor

Lesson 32: Your Digital Footprint

We will learn to: Identify, explain and apply the important ideas behind your digital footprint.



Five important ideas

Respect: means communicating without insults.

Permission: is needed before sharing another person's photo.

Digital footprint: is information left by online activity.

Report: tells a trusted adult about online harm.

Pause: means thinking before posting.

Our Ghanaian story

During online safety week, Esi had a question about your digital footprint. The group discovered that respect means communicating without insults, while permission is needed before sharing another person's photo. They drew an example and explained it to their classmates.

Try it!

Use an imaginary example or a teacher-approved offline screenshot. Discuss when to use respect and permission. Never enter a real password or share private details in this activity.

Create and share

Make a colourful, labelled poster that teaches your digital footprint. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.6.10.1.1

Lesson 32: Play & Challenge

PLAY — Choose one best answer for each question.

1. Your Digital Footprint — Choose the correct name for something that means communicating without insults.

A. Report B. Respect C. Digital footprint D. Permission

2. Your Digital Footprint — Your friend says, "This is needed before sharing another person's photo." What is the friend describing?

A. Report B. Permission C. Pause D. Digital footprint

3. Your Digital Footprint — Which idea from your digital footprint matches the clue "is information left by online activity"?

A. Permission B. Pause C. Digital footprint D. Report

4. Your Digital Footprint — The teacher asks for an example that tells a trusted adult about online harm. Which answer fits?

A. Pause B. Report C. Respect D. Permission

5. Your Digital Footprint — Can you identify the term described here: means thinking before posting?

A. Pause B. Respect C. Permission D. Digital footprint

CHALLENGE — Think and apply.

1. Your Digital Footprint — A group makes a display on your digital footprint. Their first clue is "is needed before sharing another person's photo". Which label is correct?

A. Digital footprint B. Pause C. Permission D. Report

2. Your Digital Footprint — Which answer would fit on a poster beside the words "means communicating without insults"?

A. Digital footprint B. Respect C. Permission D. Report

3. Your Digital Footprint — For a short demonstration, the teacher needs an example that is information left by online activity. Which choice would you select?

A. Digital footprint B. Pause C. Report D. Permission

4. Your Digital Footprint — During a school ICT lesson, Kofi wants to show something that means communicating without insults. Which idea should be labelled?

A. Permission B. Digital footprint C. Respect D. Report

5. Your Digital Footprint — Which concept matches the explanation "means thinking before posting" when you review your digital footprint?

A. Pause B. Respect C. Digital footprint D. Permission

6. Your Digital Footprint — A classmate confuses several ideas about your digital footprint. Which one actually is needed before sharing another person's photo?

A. Report B. Digital footprint C. Permission D. Pause

7. Your Digital Footprint — At the community learning centre, someone describes a tool or idea that means thinking before posting. Which one is it?

A. Digital footprint B. Respect C. Pause D. Permission

8. Your Digital Footprint — At home, Kofi remembers a class rule or concept that tells a trusted adult about online harm. What was it called?

A. Report B. Respect C. Permission D. Pause

9. Your Digital Footprint — Your class is solving a practical problem. One useful thing tells a trusted adult about online harm. What is its name?

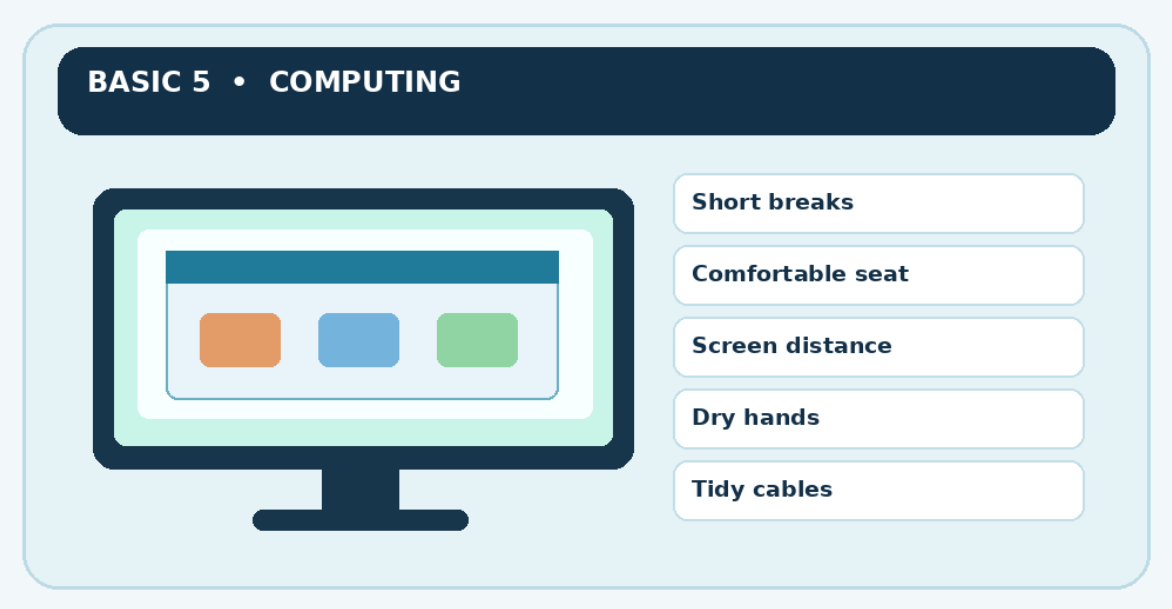
A. Report B. Respect C. Pause D. Permission

10. Your Digital Footprint — Imagine helping a younger pupil understand something that is information left by online activity. Which term would you teach?

A. Pause B. Report C. Permission D. Digital footprint

Lesson 33: Computer Health Hazards

We will learn to: Identify, explain and apply the important ideas behind computer health hazards.



Five important ideas

Short breaks: allow eyes and muscles to rest.

Comfortable seat: supports a healthy sitting position.

Screen distance: helps prevent eye strain.

Dry hands: reduce electrical risk near devices.

Tidy cables: help prevent trips and falls.

Our Ghanaian story

After a long class computer session, Yaw had a question about computer health hazards. The group discovered that short breaks allow eyes and muscles to rest, while comfortable seat supports a healthy sitting position. They drew an example and explained it to their classmates.

Try it!

Create a class safety demonstration. Show a good example of short breaks and comfortable seat and explain why the good practice matters.

Create and share

Make a colourful, labelled poster that teaches computer health hazards. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.7.1.1.1

Lesson 33: Play & Challenge

PLAY — Choose one best answer for each question.

1. Computer Health Hazards — Choose the correct name for something that allow eyes and muscles to rest.

A. Screen distance B. Short breaks C. Tidy cables D. Dry hands

2. Computer Health Hazards — Your friend says, "This supports a healthy sitting position." What is the friend describing?

A. Dry hands B. Comfortable seat C. Short breaks D. Screen distance

3. Computer Health Hazards — Which idea from computer health hazards matches the clue "helps prevent eye strain"?

A. Tidy cables B. Screen distance C. Short breaks D. Dry hands

4. Computer Health Hazards — The teacher asks for an example that reduce electrical risk near devices. Which answer fits?

A. Tidy cables B. Dry hands C. Screen distance D. Short breaks

5. Computer Health Hazards — Can you identify the term described here: help prevent trips and falls?

A. Short breaks B. Screen distance C. Dry hands D. Tidy cables

CHALLENGE — Think and apply.

1. Computer Health Hazards — A classmate confuses several ideas about computer health hazards. Which one actually supports a healthy sitting position?

A. Dry hands B. Comfortable seat C. Screen distance D. Short breaks

2. Computer Health Hazards — Which concept matches the explanation "help prevent trips and falls" when you review computer health hazards?

A. Dry hands B. Tidy cables C. Short breaks D. Screen distance

3. Computer Health Hazards — A group makes a display on computer health hazards. Their first clue is "supports a healthy sitting position". Which label is correct?

A. Dry hands B. Comfortable seat C. Screen distance D. Short breaks

4. Computer Health Hazards — Your class is solving a practical problem. One useful thing reduce electrical risk near devices. What is its name?

A. Dry hands B. Tidy cables C. Screen distance D. Short breaks

5. Computer Health Hazards — Imagine helping a younger pupil understand something that helps prevent eye strain. Which term would you teach?

A. Screen distance B. Tidy cables C. Short breaks D. Dry hands

6. Computer Health Hazards — For a short demonstration, the teacher needs an example that helps prevent eye strain. Which choice would you select?

A. Screen distance B. Short breaks C. Tidy cables D. Dry hands

7. Computer Health Hazards — At home, Akosua remembers a class rule or concept that reduce electrical risk near devices. What was it called?

A. Short breaks B. Screen distance C. Dry hands D. Tidy cables

8. Computer Health Hazards — During a school ICT lesson, Akosua wants to show something that allow eyes and muscles to rest. Which idea should be labelled?

A. Dry hands B. Tidy cables C. Screen distance D. Short breaks

9. Computer Health Hazards — Which answer would fit on a poster beside the words "allow eyes and muscles to rest"?

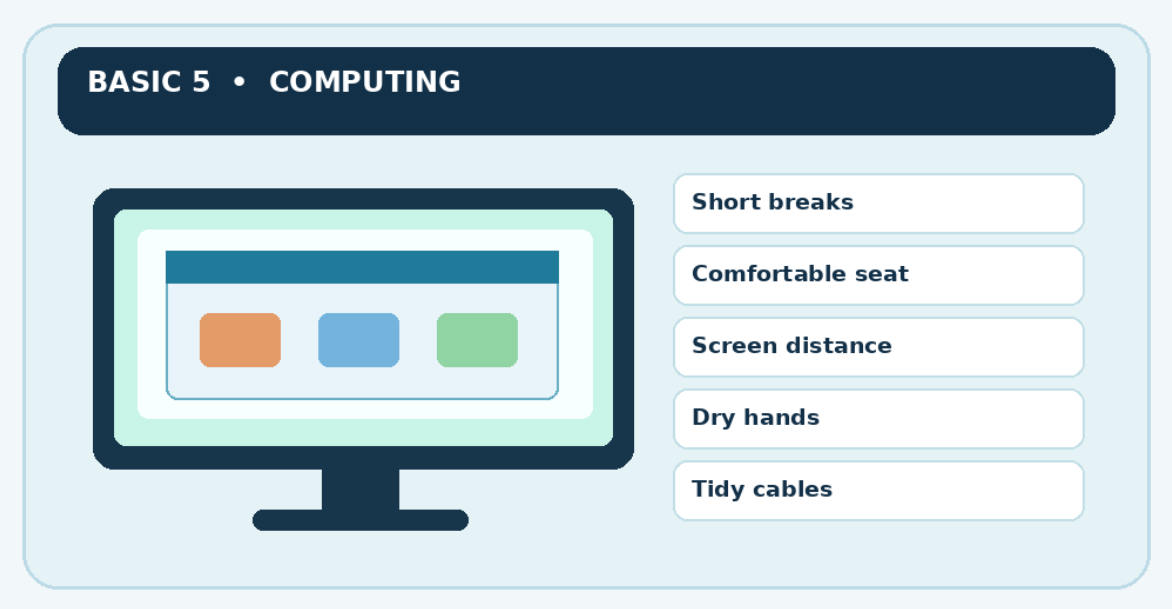
A. Dry hands B. Short breaks C. Tidy cables D. Screen distance

10. Computer Health Hazards — At the community learning centre, someone describes a tool or idea that help prevent trips and falls. Which one is it?

A. Short breaks B. Screen distance C. Tidy cables D. Dry hands

Lesson 34: Healthy Computer Posture

We will learn to: Identify, explain and apply the important ideas behind healthy computer posture.



Five important ideas

Short breaks: allow eyes and muscles to rest.

Comfortable seat: supports a healthy sitting position.

Screen distance: helps prevent eye strain.

Dry hands: reduce electrical risk near devices.

Tidy cables: help prevent trips and falls.

Our Ghanaian story

After a long class computer session, Abena had a question about healthy computer posture. The group discovered that short breaks allow eyes and muscles to rest, while comfortable seat supports a healthy sitting position. They drew an example and explained it to their classmates.

Try it!

Create a class safety demonstration. Show a good example of short breaks and comfortable seat and explain why the good practice matters.

Create and share

Make a colourful, labelled poster that teaches healthy computer posture. Include a local example and explain two of the five ideas in your own words.

Teacher mapping: B5.7.1.1.2

Lesson 34: Play & Challenge

PLAY — Choose one best answer for each question.

1. Healthy Computer Posture — Choose the correct name for something that allow eyes and muscles to rest.

A. Short breaks B. Comfortable seat C. Dry hands D. Screen distance

2. Healthy Computer Posture — Your friend says, "This supports a healthy sitting position." What is the friend describing?

A. Screen distance B. Comfortable seat C. Short breaks D. Tidy cables

3. Healthy Computer Posture — Which idea from healthy computer posture matches the clue "helps prevent eye strain"?

A. Tidy cables B. Short breaks C. Dry hands D. Screen distance

4. Healthy Computer Posture — The teacher asks for an example that reduce electrical risk near devices. Which answer fits?

A. Short breaks B. Dry hands C. Screen distance D. Comfortable seat

5. Healthy Computer Posture — Can you identify the term described here: help prevent trips and falls?

A. Comfortable seat B. Screen distance C. Short breaks D. Tidy cables

CHALLENGE — Think and apply.

1. Healthy Computer Posture — Which answer would fit on a poster beside the words "allow eyes and muscles to rest"?

A. Short breaks B. Dry hands C. Screen distance D. Comfortable seat

2. Healthy Computer Posture — For a short demonstration, the teacher needs an example that helps prevent eye strain. Which choice would you select?

A. Dry hands B. Tidy cables C. Short breaks D. Screen distance

3. Healthy Computer Posture — At the community learning centre, someone describes a tool or idea that help prevent trips and falls. Which one is it?

A. Screen distance B. Short breaks C. Comfortable seat D. Tidy cables

4. Healthy Computer Posture — Your class is solving a practical problem. One useful thing reduce electrical risk near devices. What is its name?

A. Dry hands B. Short breaks C. Screen distance D. Comfortable seat

5. Healthy Computer Posture — Imagine helping a younger pupil understand something that helps prevent eye strain. Which term would you teach?

A. Short breaks B. Tidy cables C. Screen distance D. Dry hands

6. Healthy Computer Posture — A classmate confuses several ideas about healthy computer posture. Which one actually supports a healthy sitting position?

A. Tidy cables B. Short breaks C. Screen distance D. Comfortable seat

7. Healthy Computer Posture — Which concept matches the explanation "help prevent trips and falls" when you review healthy computer posture?

A. Comfortable seat B. Tidy cables C. Screen distance D. Short breaks

8. Healthy Computer Posture — A group makes a display on healthy computer posture. Their first clue is "supports a healthy sitting position". Which label is correct?

A. Tidy cables B. Screen distance C. Short breaks D. Comfortable seat

9. Healthy Computer Posture — At home, Esi remembers a class rule or concept that reduce electrical risk near devices. What was it called?

A. Screen distance B. Short breaks C. Comfortable seat D. Dry hands

10. Healthy Computer Posture — During a school ICT lesson, Esi wants to show something that allow eyes and muscles to rest. Which idea should be labelled?

A. Screen distance B. Comfortable seat C. Short breaks D. Dry hands

Teacher answer guide

Review the correct term and ask each learner to explain why it fits.

Lesson 1 — Recognising Output Devices

Play: 1. Monitor | 2. Printer | 3. Speaker | 4. Projector | 5. Headphones

Challenge: 1. Printer | 2. Projector | 3. Speaker | 4. Speaker | 5. Printer | 6. Monitor | 7. Monitor | 8. Projector | 9. Headphones | 10. Headphones

Lesson 2 — Uses of Output Devices

Play: 1. Monitor | 2. Printer | 3. Speaker | 4. Projector | 5. Headphones

Challenge: 1. Monitor | 2. Printer | 3. Projector | 4. Speaker | 5. Headphones | 6. Printer | 7. Projector | 8. Headphones | 9. Speaker | 10. Monitor

Lesson 3 — Analogue and Digital Devices

Play: 1. Input device | 2. Output device | 3. Storage device | 4. Processor | 5. Peripheral

Challenge: 1. Storage device | 2. Input device | 3. Processor | 4. Input device | 5. Output device | 6. Processor | 7. Storage device | 8. Output device | 9. Peripheral | 10. Peripheral

Lesson 4 — Advanced Mouse Practice

Play: 1. Single click | 2. Double click | 3. Drag | 4. Scroll wheel | 5. Right click

Challenge: 1. Drag | 2. Double click | 3. Right click | 4. Drag | 5. Single click | 6. Scroll wheel | 7. Scroll wheel | 8. Right click | 9. Single click | 10. Double click

Lesson 5 — Keyboard Rows and Short Text

Play: 1. Home row | 2. Top row | 3. Bottom row | 4. Practice | 5. Posture

Challenge: 1. Top row | 2. Bottom row | 3. Posture | 4. Practice | 5. Top row | 6. Posture | 7. Practice | 8. Home row | 9. Home row | 10. Bottom row

Lesson 6 — Windows Desktop Interface

Play: 1. Desktop | 2. Icon | 3. Taskbar | 4. Wallpaper | 5. Start menu

Challenge: 1. Taskbar | 2. Start menu | 3. Wallpaper | 4. Taskbar | 5. Desktop | 6. Icon | 7. Start menu | 8. Icon | 9. Desktop | 10. Wallpaper

Lesson 7 — How Computers Have Changed

Play: 1. Vacuum tube | 2. Transistor | 3. Integrated circuit | 4. Microprocessor | 5. Personal computer

Challenge: 1. Integrated circuit | 2. Personal computer | 3. Personal computer | 4. Microprocessor | 5. Vacuum tube | 6. Transistor | 7. Transistor | 8. Integrated circuit | 9. Microprocessor | 10. Vacuum tube

Lesson 8 — Create a Desktop Background

Play: 1. Title | 2. Font | 3. Colour | 4. Picture | 5. Layout

Challenge: 1. Colour | 2. Title | 3. Font | 4. Picture | 5. Picture | 6. Font | 7. Layout | 8. Layout | 9. Colour | 10. Title

Lesson 9 — Start Menu and Pinned Apps

Play: 1. Desktop | 2. Icon | 3. Taskbar | 4. Wallpaper | 5. Start menu

Challenge: 1. Icon | 2. Wallpaper | 3. Icon | 4. Desktop | 5. Desktop | 6. Start menu | 7. Taskbar | 8. Wallpaper | 9. Start menu | 10. Taskbar

Lesson 10 — Desktop Shortcuts

Play: 1. File Explorer | 2. Navigation pane | 3. Search box | 4. Storage drive | 5. Recycle Bin

Challenge: 1. Navigation pane | 2. Search box | 3. Storage drive | 4. File Explorer | 5. Recycle Bin | 6. Storage drive | 7. Navigation pane | 8. File Explorer | 9. Recycle Bin | 10. Search box

Lesson 11 — Move and Copy Files

Play: 1. Folder | 2. File name | 3. Copy | 4. Move | 5. Rename

Challenge: 1. Rename | 2. Move | 3. File name | 4. Copy | 5. Rename | 6. File name | 7. Folder | 8. Move | 9. Copy | 10. Folder

Lesson 12 — Use File Explorer

Play: 1. File Explorer | 2. Navigation pane | 3. Search box | 4. Storage drive | 5. Recycle Bin

Challenge: 1. Storage drive | 2. Recycle Bin | 3. Storage drive | 4. Search box | 5. File Explorer | 6. Navigation pane | 7. Recycle Bin | 8. Navigation pane | 9. Search box | 10. File Explorer

Lesson 13 — Data From Experiments

Play: 1. Data | 2. Information | 3. Survey | 4. Observation | 5. Table

Challenge: 1. Survey | 2. Survey | 3. Table | 4. Observation | 5. Data | 6. Observation | 7. Data | 8. Information | 9. Table | 10. Information

Lesson 14 — Primary Sources of Data

Play: 1. Interview | 2. Photograph | 3. Newspaper | 4. Questionnaire | 5. Recording

Challenge: 1. Questionnaire | 2. Photograph | 3. Newspaper | 4. Recording | 5. Interview | 6. Recording | 7. Photograph | 8. Interview | 9. Questionnaire | 10. Newspaper

Lesson 15 — Collect a Data Sample

Play: 1. Research question | 2. Sample | 3. Consent | 4. Tally | 5. Conclusion

Challenge: 1. Conclusion | 2. Research question | 3. Sample | 4. Conclusion | 5. Consent | 6. Tally | 7. Tally | 8. Consent | 9. Sample | 10. Research question

Lesson 16 — Analyse Charts and Tables

Play: 1. Count | 2. Sum | 3. Average | 4. Maximum | 5. Minimum

Challenge: 1. Minimum | 2. Average | 3. Maximum | 4. Count | 5. Maximum | 6. Count | 7. Sum | 8. Sum | 9. Average | 10. Minimum

Lesson 17 — How Technology Affects Communities

Play: 1. Benefit | 2. Risk | 3. Healthcare | 4. Farming | 5. Privacy

Challenge: 1. Farming | 2. Healthcare | 3. Risk | 4. Privacy | 5. Benefit | 6. Farming | 7. Healthcare | 8. Risk | 9. Privacy | 10. Benefit

Lesson 18 — Hardware, Software and Networks

Play: 1. Input device | 2. Output device | 3. Storage device | 4. Processor | 5. Peripheral

Challenge: 1. Peripheral | 2. Storage device | 3. Processor | 4. Storage device | 5. Output device | 6. Input device | 7. Input device | 8. Output device | 9. Processor | 10. Peripheral

Lesson 19 — PowerPoint Design and Animation

Play: 1. Slide | 2. Theme | 3. Transition | 4. Animation | 5. Presenter

Challenge: 1. Slide | 2. Theme | 3. Presenter | 4. Animation | 5. Transition | 6. Presenter | 7. Slide | 8. Theme | 9. Animation | 10. Transition

Lesson 20 — Present Five Slides

Play: 1. Opening slide | 2. Main slide | 3. Image | 4. Closing slide | 5. Rehearsal

Challenge: 1. Opening slide | 2. Closing slide | 3. Image | 4. Rehearsal | 5. Rehearsal | 6. Image | 7. Main slide | 8. Opening slide | 9. Main slide | 10. Closing slide

Lesson 21 — Word Processing Ribbons

Play: 1. Word processor | 2. Bold | 3. Alignment | 4. Clipboard | 5. Spell check

Challenge: 1. Clipboard | 2. Word processor | 3. Bold | 4. Clipboard | 5. Word processor | 6. Spell check | 7. Spell check | 8. Alignment | 9. Alignment | 10. Bold

Lesson 22 — Design a Well-Formatted Document

Play: 1. Title | 2. Font | 3. Colour | 4. Picture | 5. Layout

Challenge: 1. Title | 2. Title | 3. Colour | 4. Layout | 5. Font | 6. Layout | 7. Font | 8. Picture | 9. Picture | 10. Colour

Lesson 23 — Introducing Databases

Play: 1. Database | 2. Record | 3. Field | 4. Table | 5. Query

Challenge: 1. Table | 2. Field | 3. Table | 4. Database | 5. Query | 6. Database | 7. Record | 8. Field | 9. Record | 10. Query

Lesson 24 — Why We Use Databases

Play: 1. Database | 2. Record | 3. Field | 4. Table | 5. Query

Challenge: 1. Query | 2. Record | 3. Database | 4. Field | 5. Table | 6. Database | 7. Field | 8. Record | 9. Query | 10. Table

Lesson 25 — Database Tables and Records

Play: 1. Database | 2. Record | 3. Field | 4. Table | 5. Query

Challenge: 1. Table | 2. Query | 3. Database | 4. Record | 5. Table | 6. Field | 7. Record | 8. Query | 9. Field | 10. Database

Lesson 26 — Programming Languages

Play: 1. Algorithm | 2. Sequence | 3. Repeat | 4. Condition | 5. Debugging

Challenge: 1. Debugging | 2. Algorithm | 3. Sequence | 4. Debugging | 5. Algorithm | 6. Repeat | 7. Condition | 8. Sequence | 9. Repeat | 10. Condition

Lesson 27 — Understanding Web Browsers

Play: 1. Web browser | 2. Address bar | 3. Home page | 4. Hyperlink | 5. Back button

Challenge: 1. Web browser | 2. Address bar | 3. Web browser | 4. Home page | 5. Hyperlink | 6. Back button | 7. Home page | 8. Hyperlink | 9. Back button | 10. Address bar

Lesson 28 — What Is a Web Page?

Play: 1. Web browser | 2. Address bar | 3. Home page | 4. Hyperlink | 5. Back button

Challenge: 1. Hyperlink | 2. Address bar | 3. Address bar | 4. Hyperlink | 5. Back button | 6. Home page | 7. Web browser | 8. Web browser | 9. Home page | 10. Back button

Lesson 29 — Following Hyperlinks

Play: 1. Web browser | 2. Address bar | 3. Home page | 4. Hyperlink | 5. Back button

Challenge: 1. Address bar | 2. Web browser | 3. Home page | 4. Back button | 5. Hyperlink | 6. Web browser | 7. Back button | 8. Home page | 9. Address bar | 10. Hyperlink

Lesson 30 — Email Services

Play: 1. Recipient | 2. Subject | 3. Message body | 4. Attachment | 5. Reply

Challenge: 1. Subject | 2. Reply | 3. Message body | 4. Recipient | 5. Reply | 6. Attachment | 7. Message body | 8. Subject | 9. Recipient | 10. Attachment

Lesson 31 — Internet of Things

Play: 1. Sensor | 2. Smart device | 3. Network | 4. Automated irrigation | 5. Security

Challenge: 1. Security | 2. Security | 3. Network | 4. Sensor | 5. Smart device | 6. Automated irrigation | 7. Smart device | 8. Sensor | 9. Network | 10. Automated irrigation

Lesson 32 — Your Digital Footprint

Play: 1. Respect | 2. Permission | 3. Digital footprint | 4. Report | 5. Pause

Challenge: 1. Permission | 2. Respect | 3. Digital footprint | 4. Respect | 5. Pause | 6. Permission | 7. Pause | 8. Report | 9. Report | 10. Digital footprint

Lesson 33 — Computer Health Hazards

Play: 1. Short breaks | 2. Comfortable seat | 3. Screen distance | 4. Dry hands | 5. Tidy cables

Challenge: 1. Comfortable seat | 2. Tidy cables | 3. Comfortable seat | 4. Dry hands | 5. Screen distance | 6. Screen distance | 7. Dry hands | 8. Short breaks | 9. Short breaks | 10. Tidy cables

Lesson 34 — Healthy Computer Posture

Play: 1. Short breaks | 2. Comfortable seat | 3. Screen distance | 4. Dry hands | 5. Tidy cables

Challenge: 1. Short breaks | 2. Screen distance | 3. Tidy cables | 4. Dry hands | 5. Screen distance | 6. Comfortable seat | 7. Tidy cables | 8. Comfortable seat | 9. Dry hands | 10. Short breaks